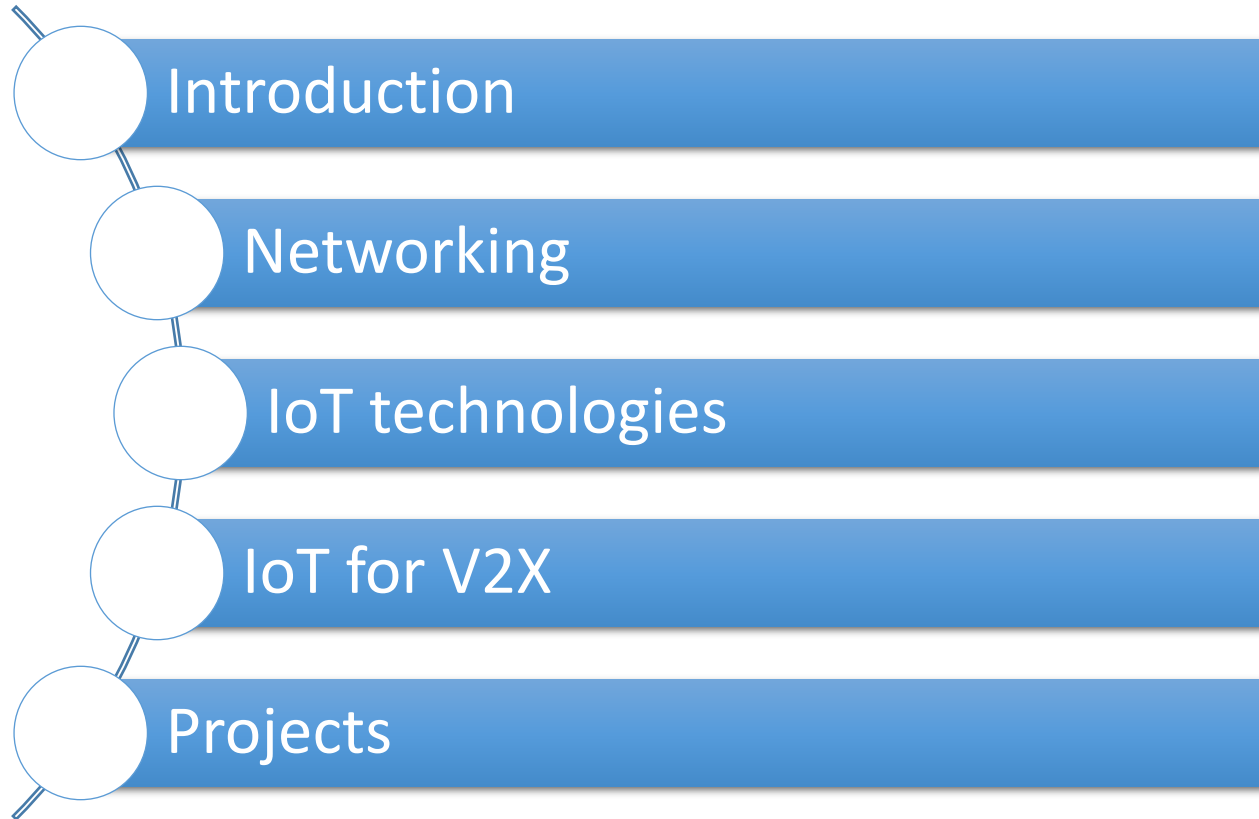


Internet Of Things

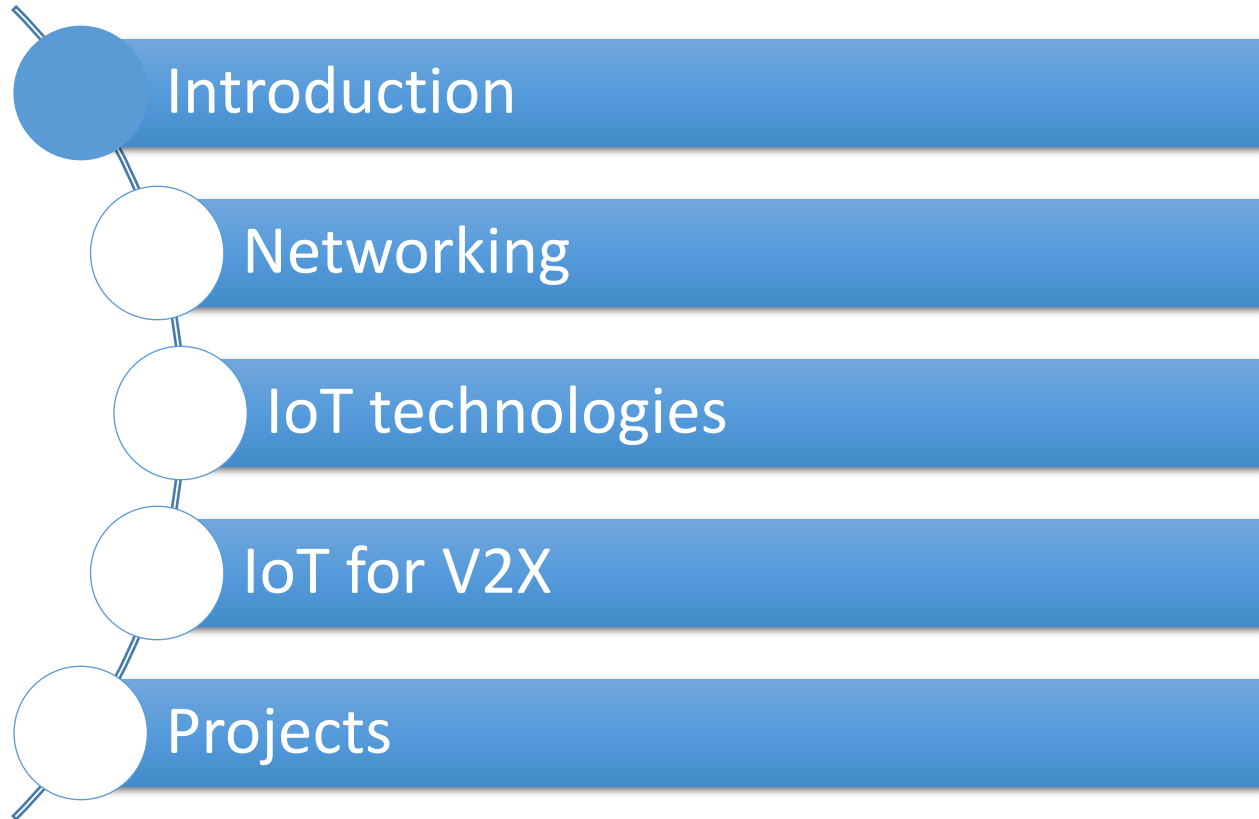
Mehdi MCHAREK, Dr.-Eng.

e-mail : mehdi.mcharek@gmail.com

Agenda



Agenda



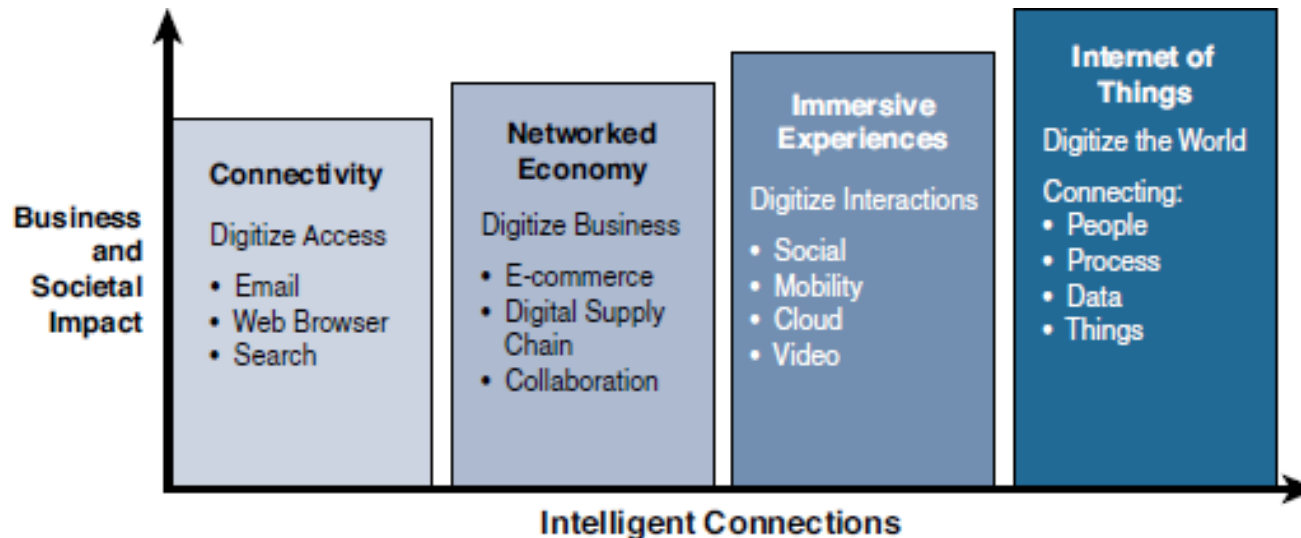
IoT definition

“Imagine a world where just about anything you can think of is online and communicating to other things and people in order to enable new services that enhance our lives. From self-driving drones delivering your grocery order to sensors in your clothing monitoring your health, the world you know is set to undergo a major technological shift forward.

This shift is known collectively as the Internet of Things (IoT).” *CISCO Report*

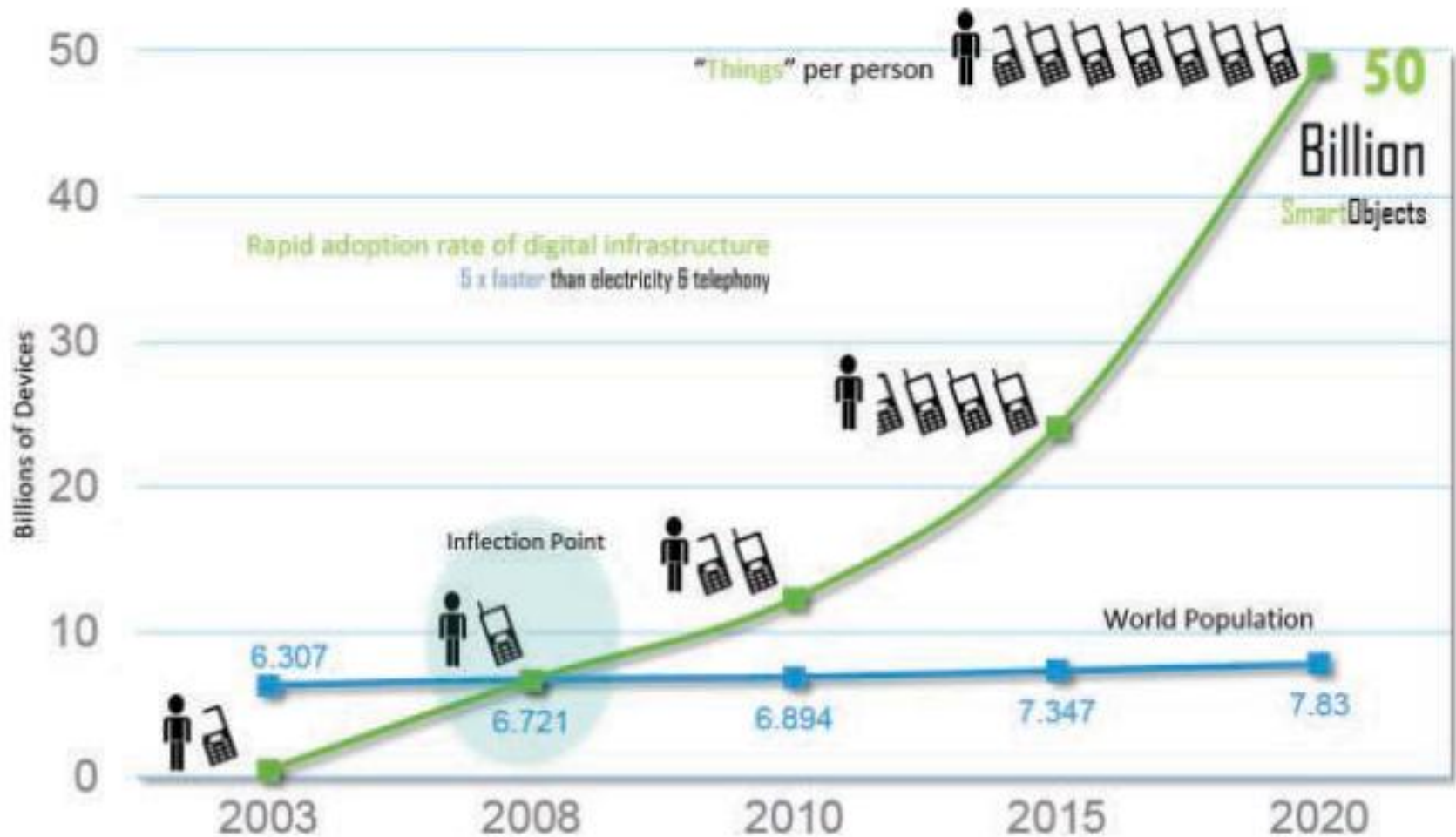
Definition : *The Internet of things (IoT)* is the network of physical objects accessed through the Internet.

Internet evolution

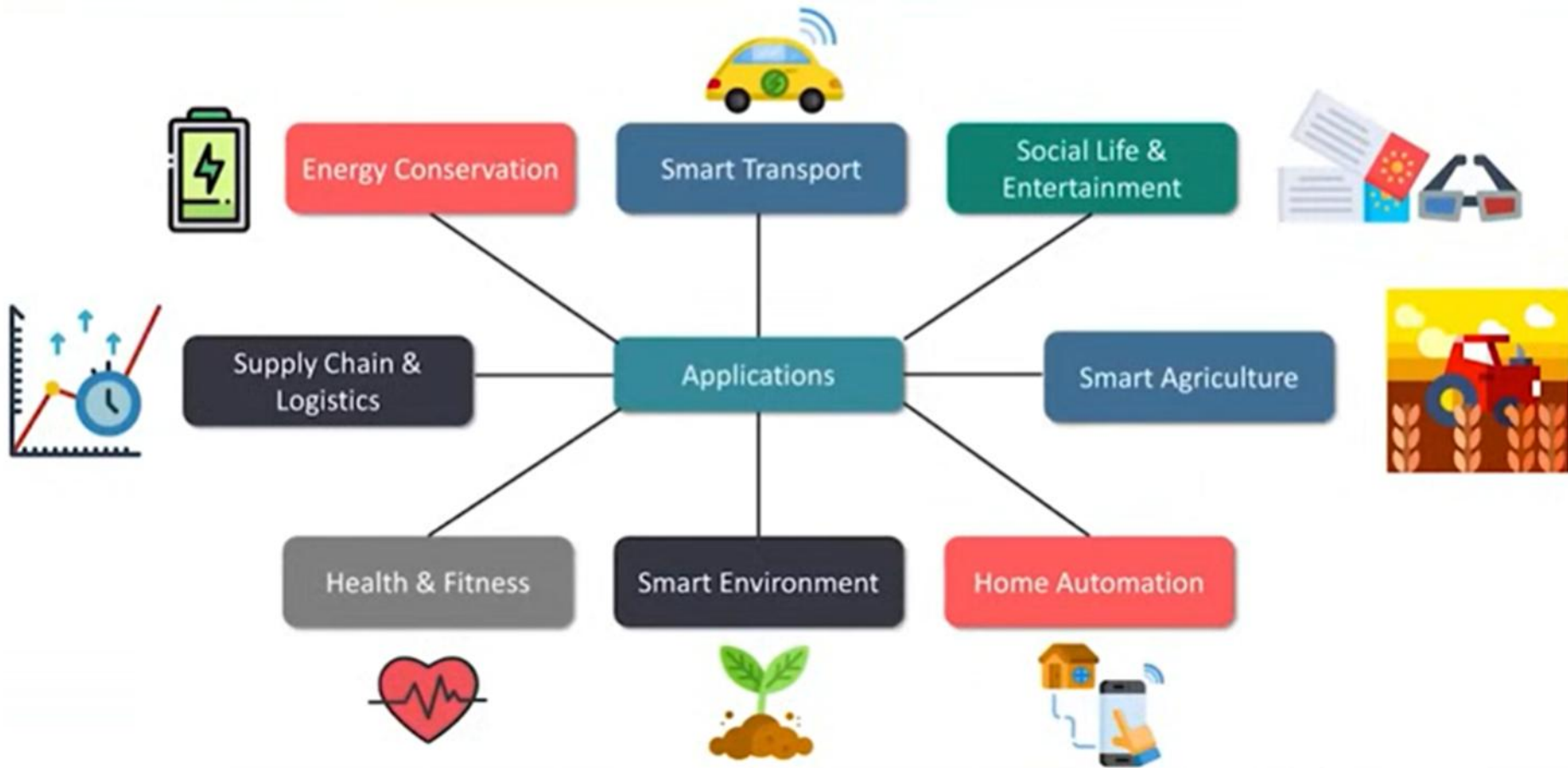


Internet Phase	Definition
Connectivity (Digitize access)	This phase connected people to email, web services, and search so that information is easily accessed.
Networked Economy (Digitize business)	This phase enabled e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes.
Immersive Experiences (Digitize interactions)	This phase extended the Internet experience to encompass widespread video and social media while always being connected through mobility. More and more applications are moved into the cloud.
Internet of Things (Digitize the world)	This phase is adding connectivity to objects and machines in the world around us to enable new services and experiences. It is connecting the unconnected.

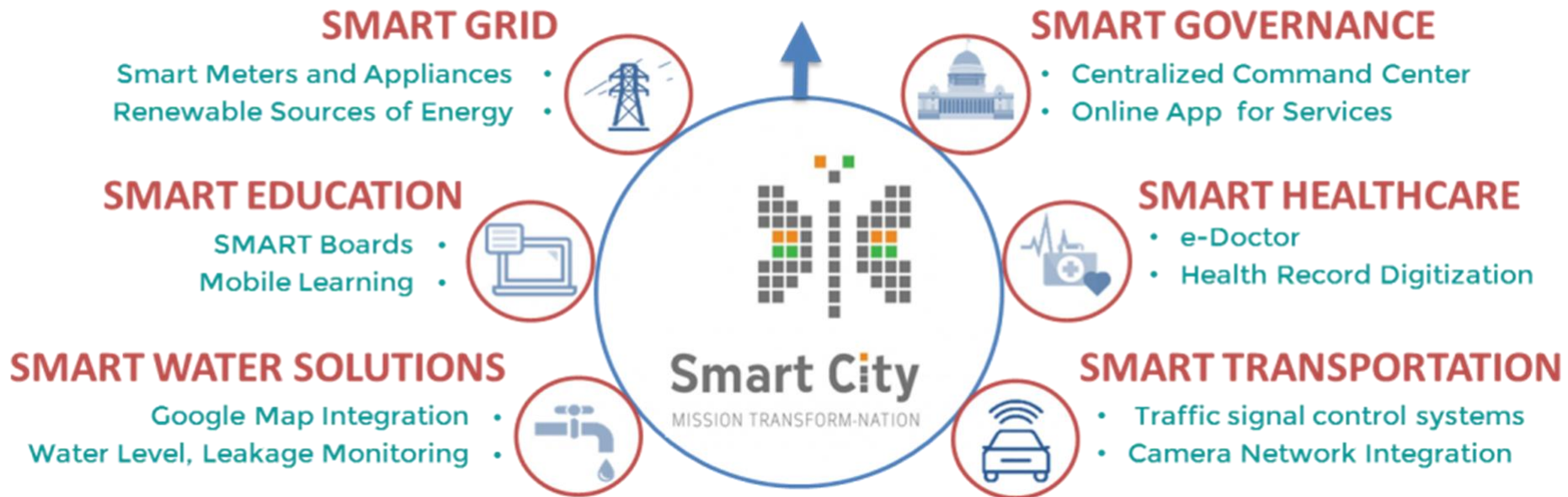
The rapid growth of connected devices



IoT applications



Smart City



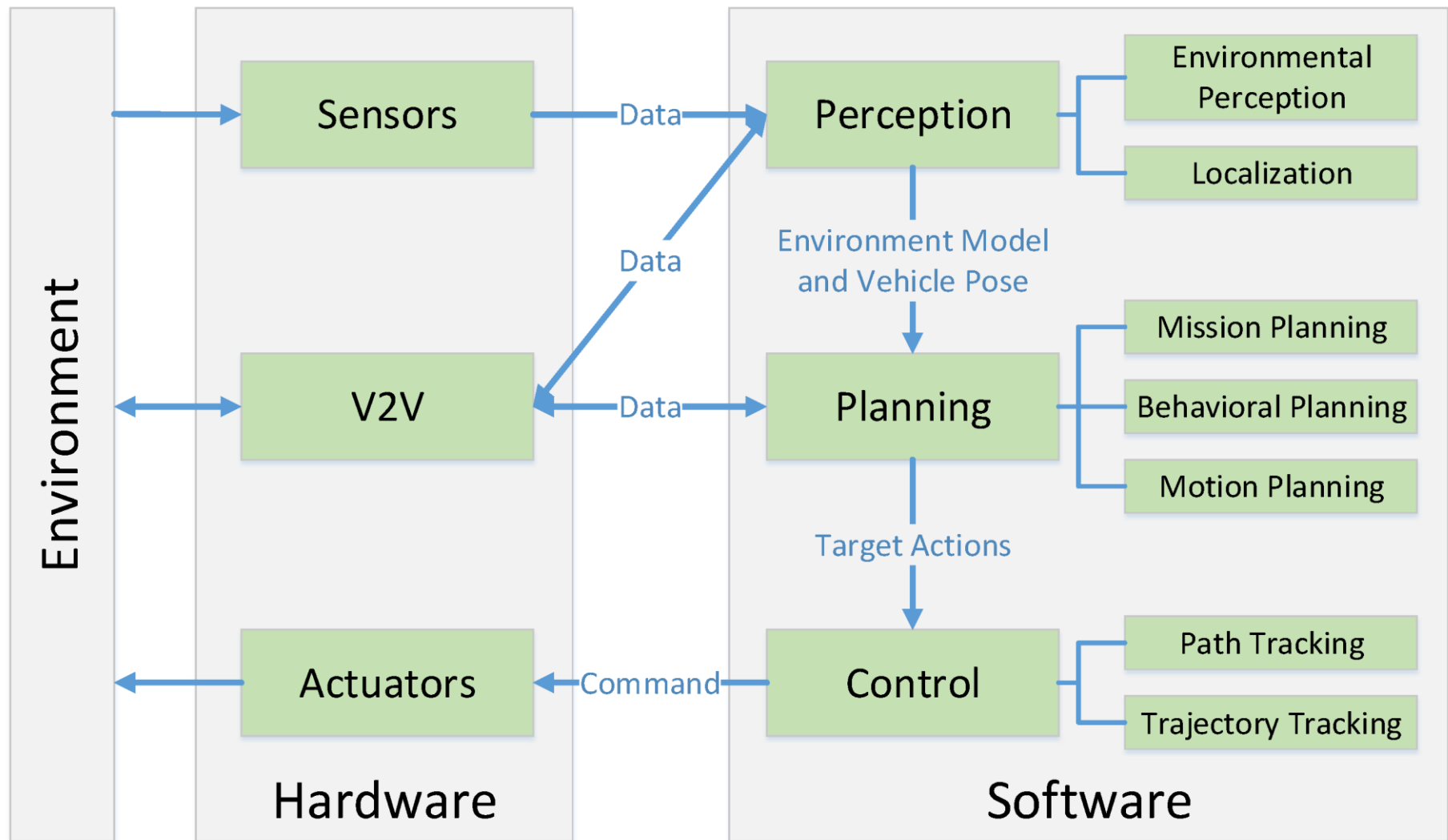
Connected roadways

IoT is going to allow self-driving vehicles to better interact with the transportation system around them through bidirectional data exchanges while also providing important data to the riders. Self-driving vehicles need always-on, reliable communications and data from other transportation-related sensors to reach their full potential.

Connected roadways is the term associated with both the driver and driverless cars fully integrating with the surrounding transportation infrastructure



Architecture of an autonomous system

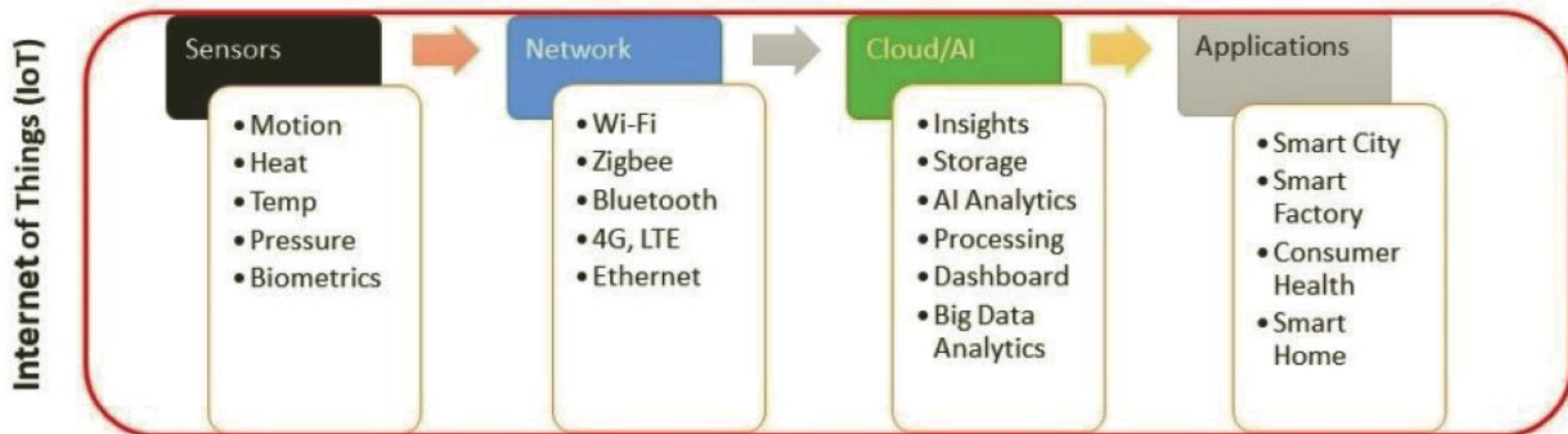


Roadways challenges

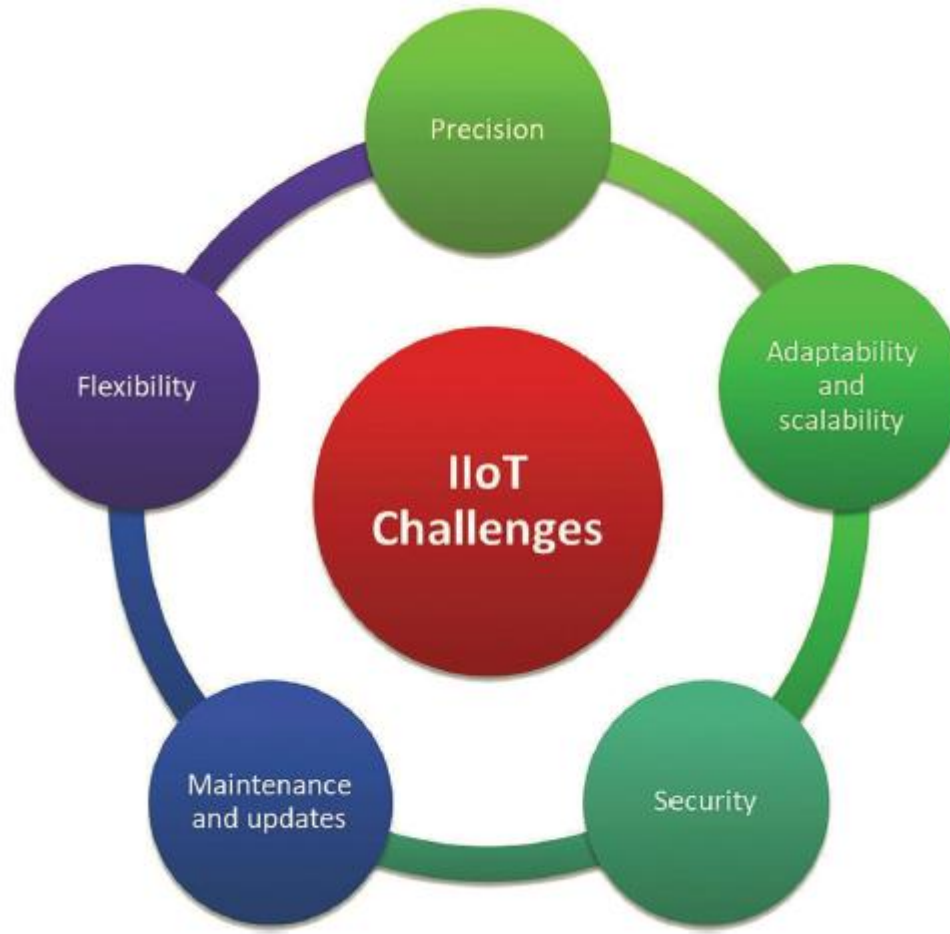
Challenge	Supporting Data
Safety	According to the US Department of Transportation, 5.6 million crashes were reported in 2012 alone, resulting in more than 33,000 fatalities. IoT and the enablement of connected vehicle technologies will empower drivers with the tools they need to anticipate potential crashes and significantly reduce the number of lives lost each year.
Mobility	More than a billion cars are on the roads worldwide. Connected vehicle mobility applications can enable system operators and drivers to make more informed decisions, which can, in turn, reduce travel delays. Congestion causes 5.5 billion hours of travel delay per year, and reducing travel delays is more critical than ever before. In addition, communication between mass transit, emergency response vehicles, and traffic management infrastructures help optimize the routing of vehicles, further reducing potential delays.
Environment	According to the American Public Transportation Association, each year transit systems can collectively reduce carbon dioxide (CO ₂) emissions by 16.2 million metric tons by reducing private vehicle miles. Connected vehicle environmental applications will give all travelers the real-time information they need to make “green” transportation choices.

Sources: Traffic Safety Facts, 2010; National Highway Traffic Safety Administration, June 2012; and WHO Global Status Report on Road Safety, 2013.

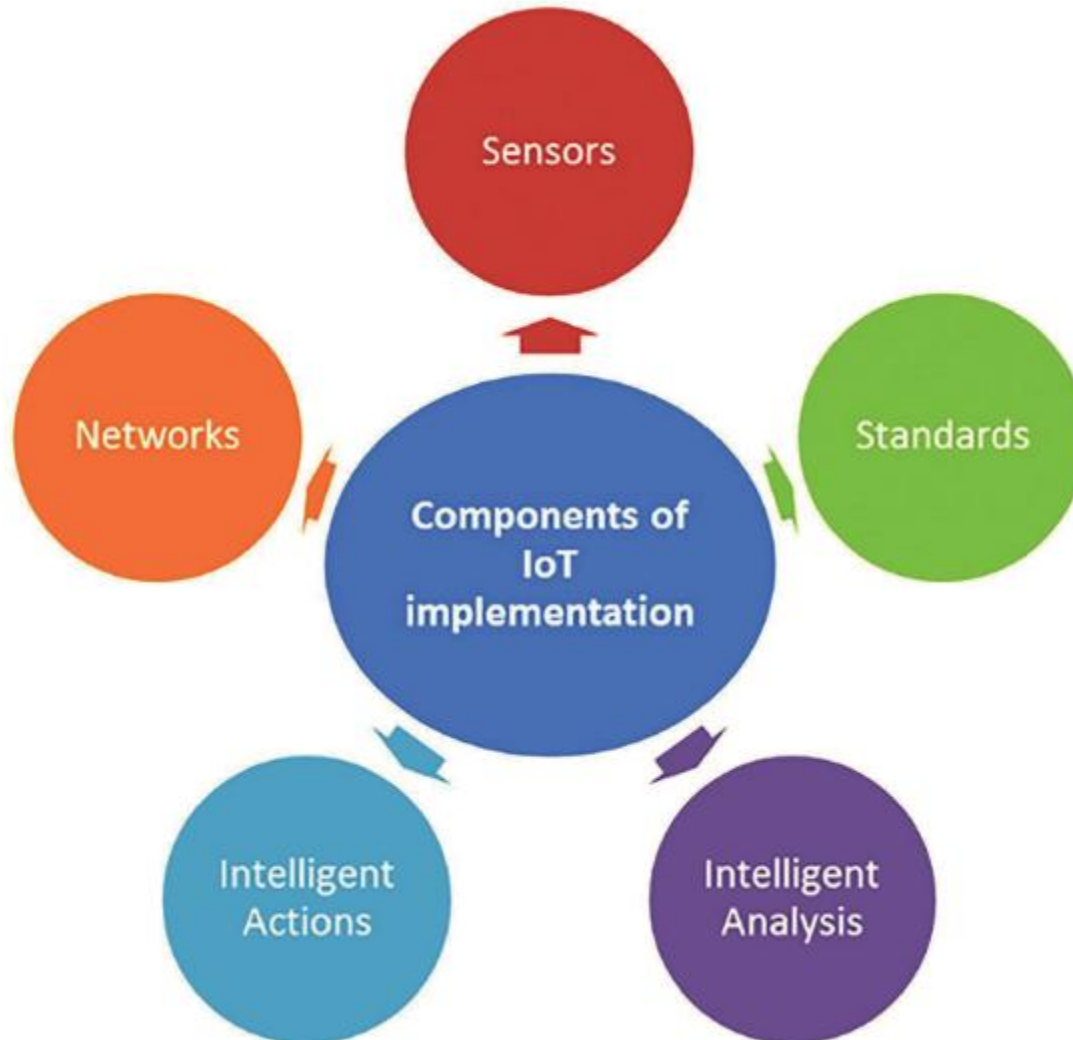
IoT components



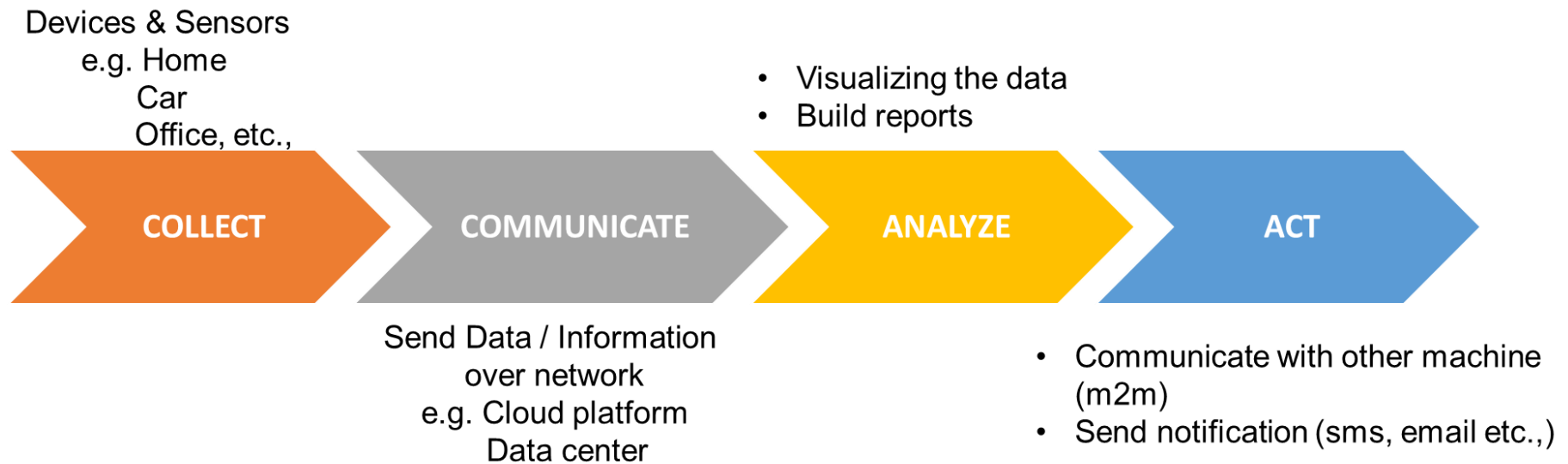
IoT challenges



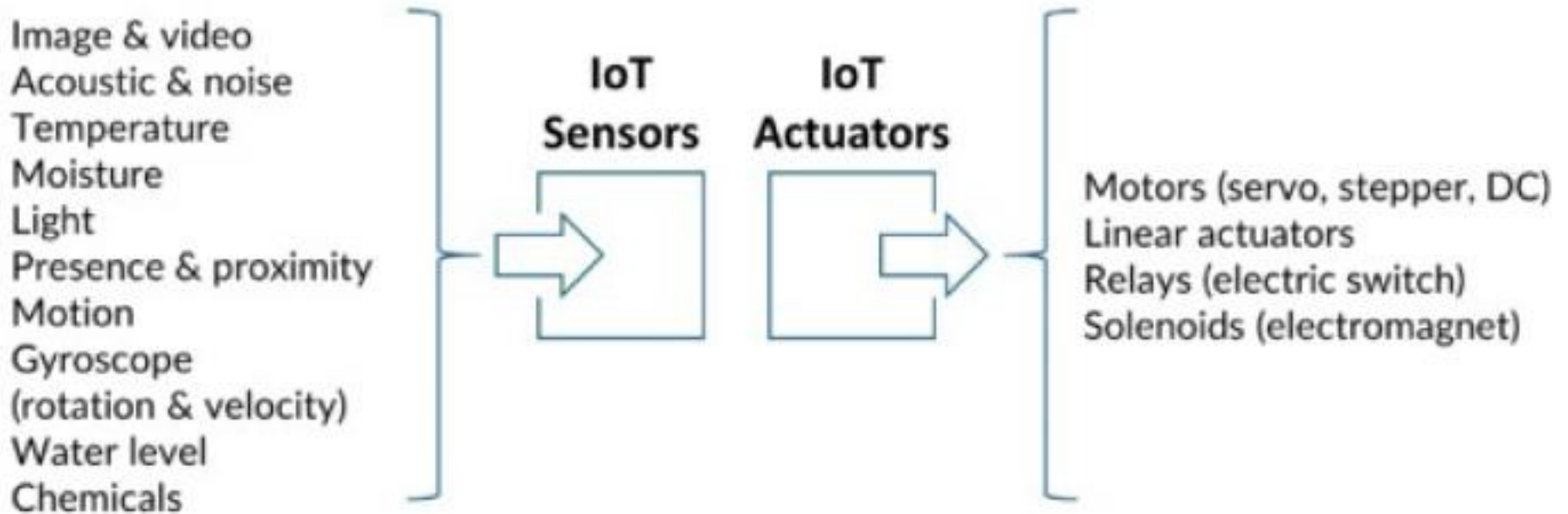
Components of IoT implementation



IoT Lifecycle



IoT sensors and actuators



Example : car connected services

DECouvrez NOS SERVICES CONNECTÉS



CONNECT ONE : VOTRE ASSISTANT INTÉGRÉ

Prenez le contrôle total de votre véhicule et de votre expérience de conduite à l'aide de votre smartphone.

INCLUS dans les TSCV

DECouvrez



CONNECT PLUS : EXPLOREZ DE NOUVEAUX HORIZONS*

Donnez le contrôle total de votre expérience de conduite à l'aide de votre smartphone.

INCLUS dans les TSCV

DECouvrez



TRAFIC EN TEMPS RÉEL

Consultez les données en temps réel de votre véhicule et de votre expérience de conduite à l'aide de votre smartphone.

INCLUS dans les TSCV

DECouvrez



ZONES DE DANGER

Consultez les zones de danger de votre véhicule et de votre expérience de conduite à l'aide de votre smartphone.

A DÉCLANCHER à l'aide de votre smartphone

DECouvrez



PACK NAVIGATION CONNECTÉE

Consultez les données de votre véhicule et de votre expérience de conduite à l'aide de votre smartphone.

INCLUS dans les TSCV

DECouvrez

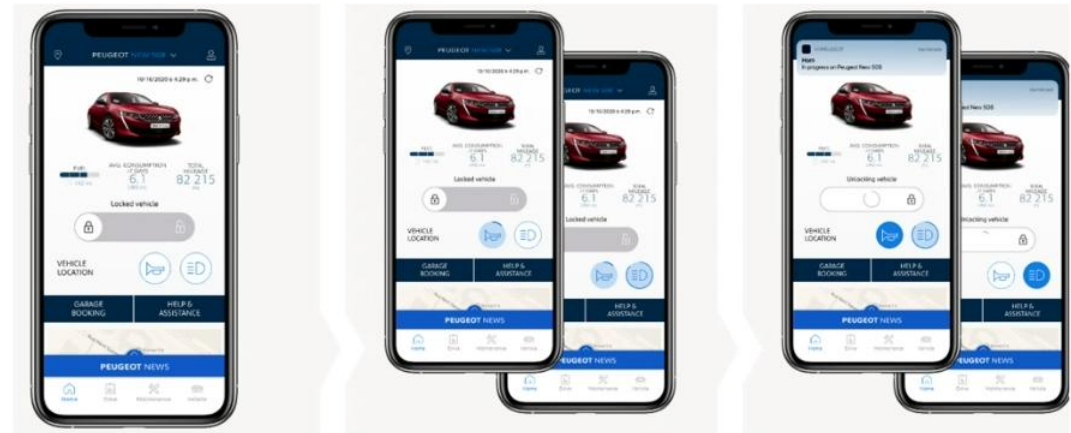


TELEMANTENANCE

Consultez les données de votre véhicule et de votre expérience de conduite à l'aide de votre smartphone.

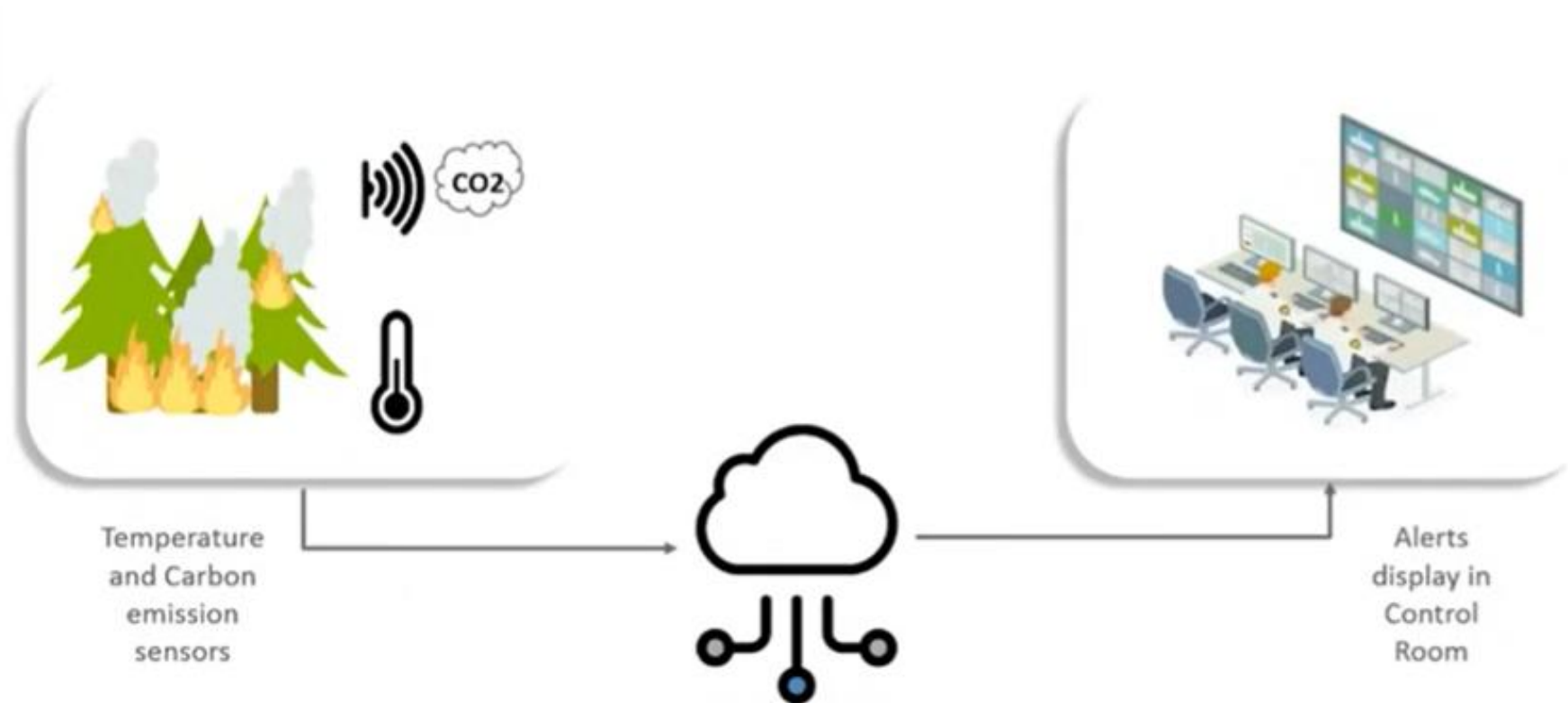
GRATUIT à l'aide de votre smartphone

DECouvrez

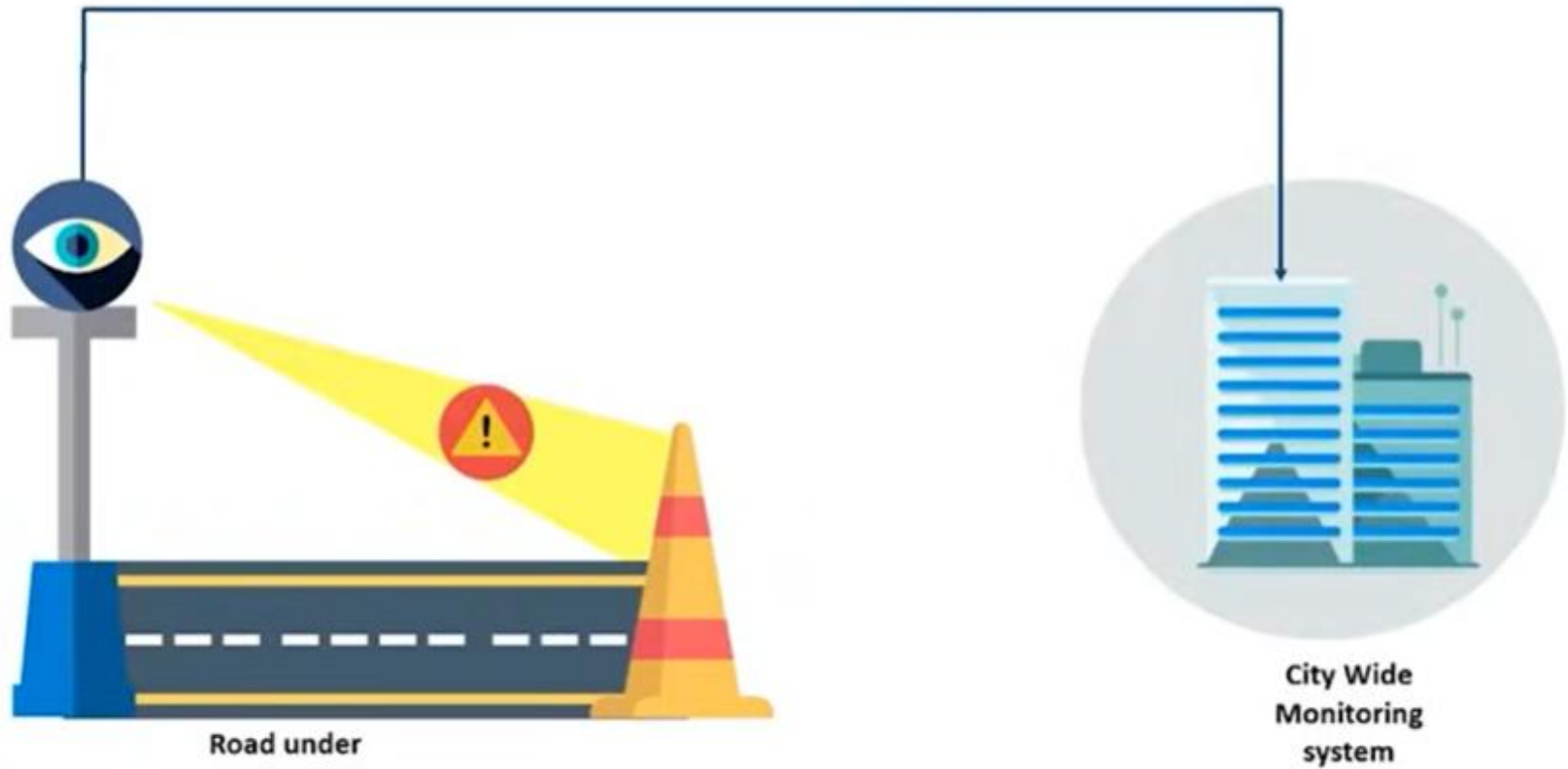


Remote access permet de consulter des informations sur le véhicule et de déclencher des actions à distance (ouvrir/fermer les portes par ex). Elle permet aussi de voir le statut du véhicule. Tout cela via votre smartphone.

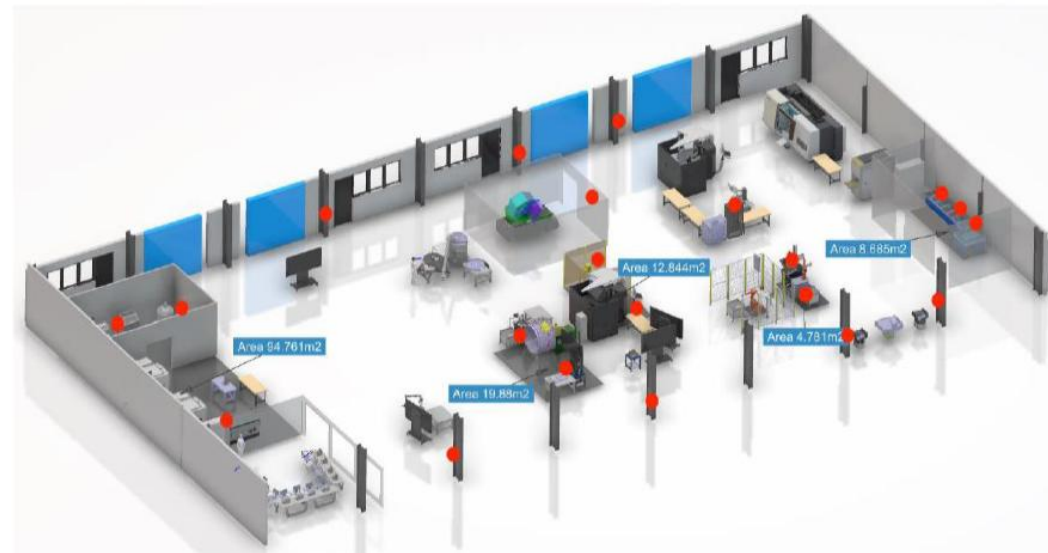
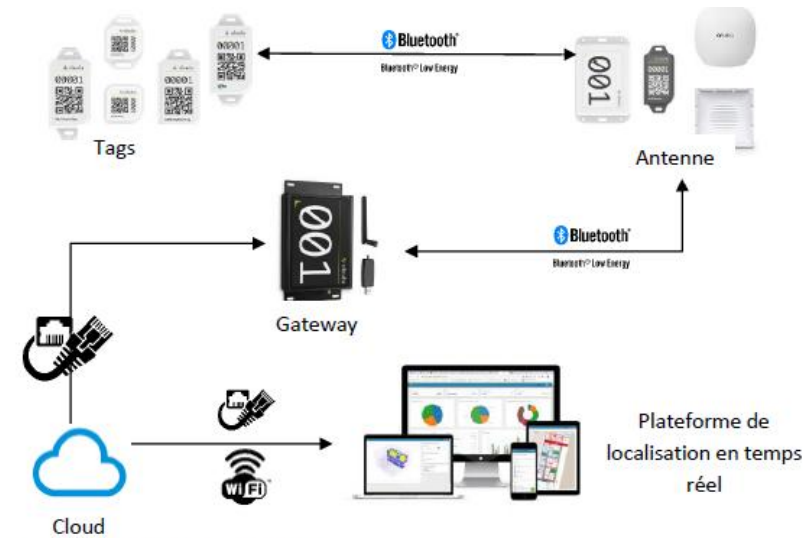
Example : forest fire alert system



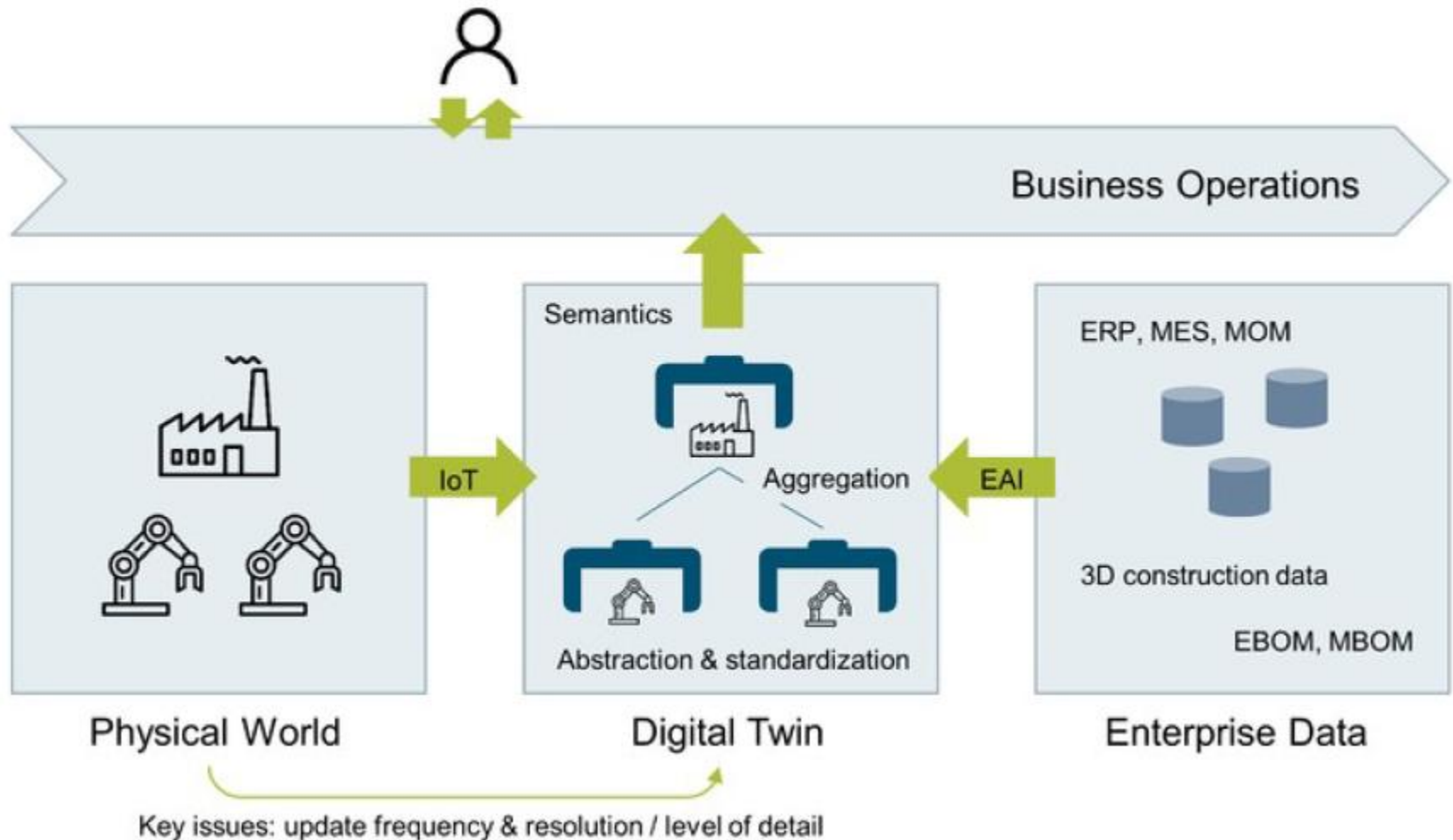
Example : smart traffic system



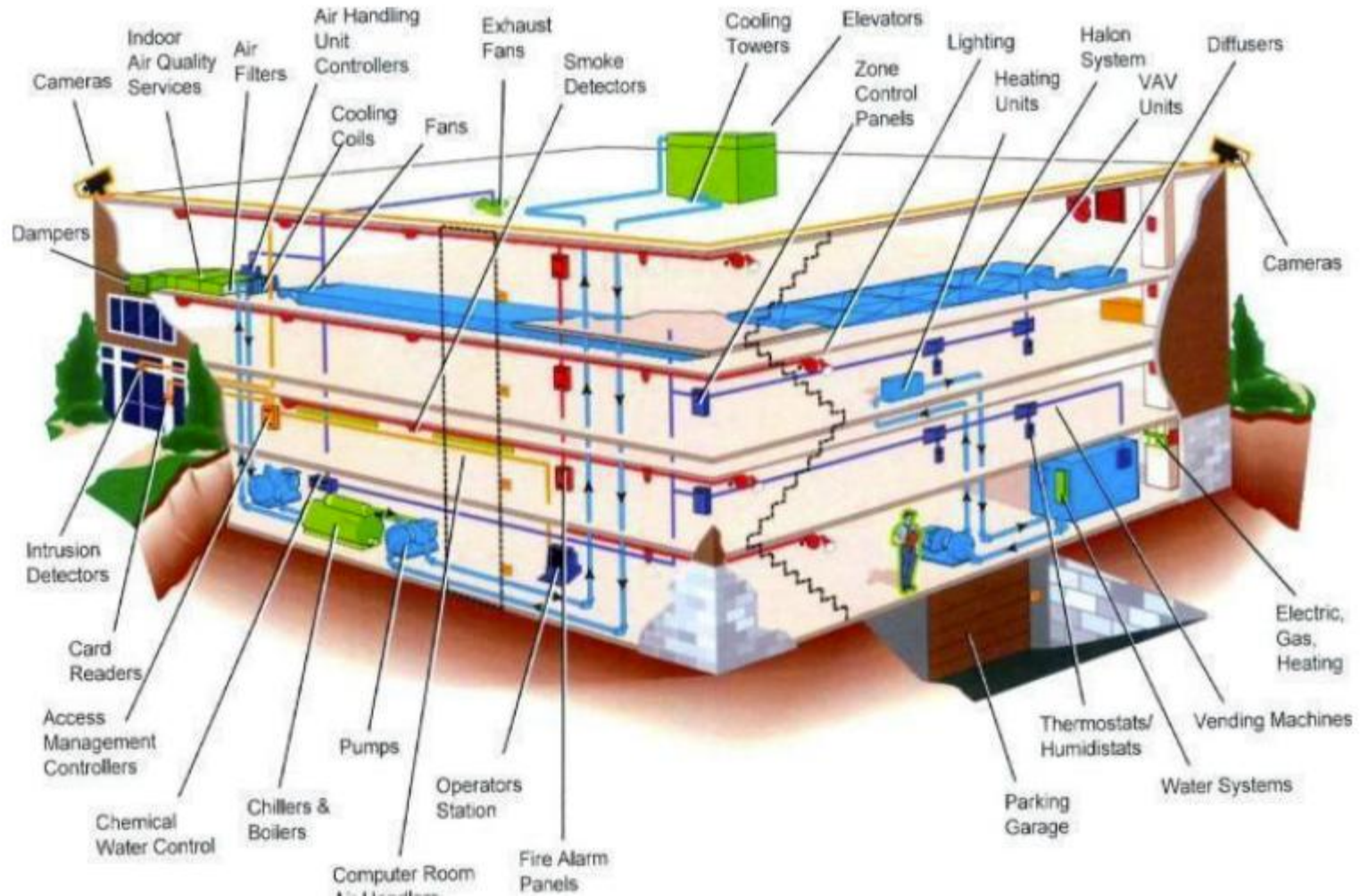
Example : Real time localisation of tools



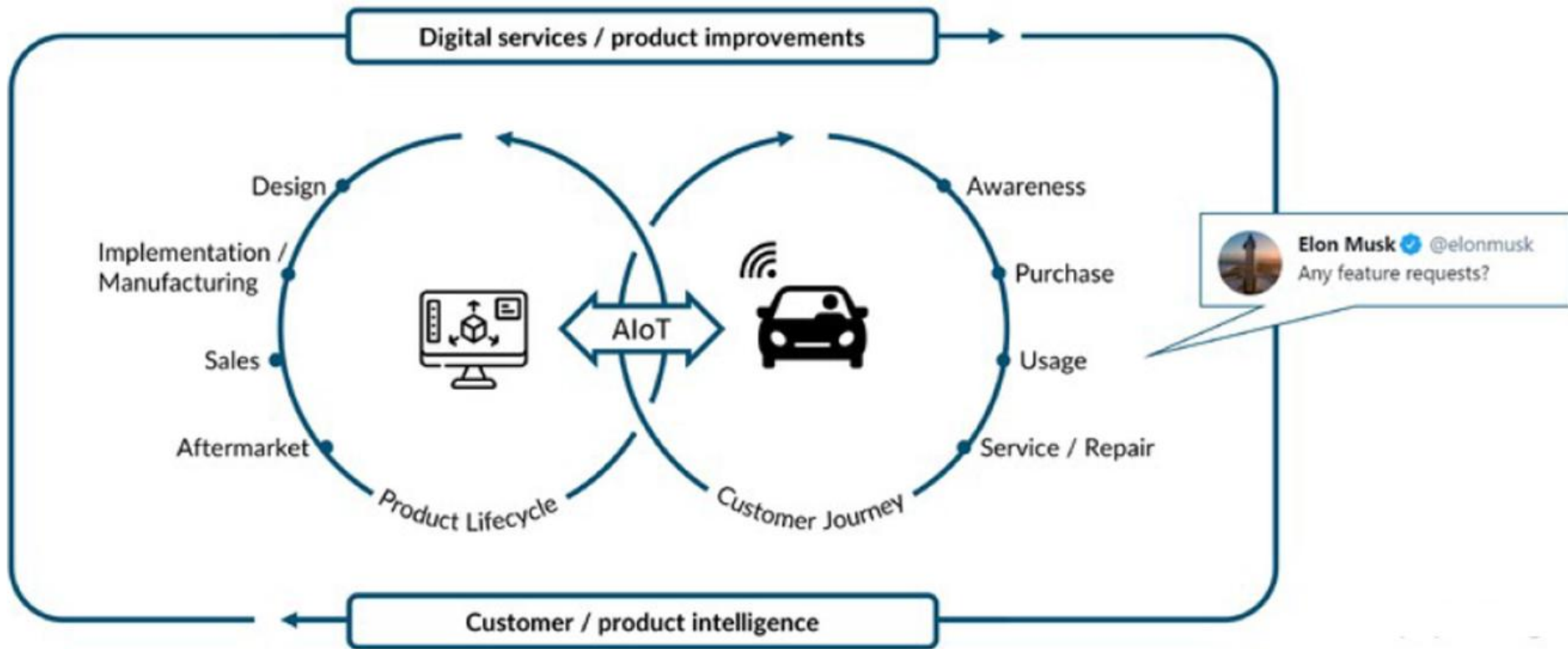
Example : IoT Digital Twin



Example : Smart home



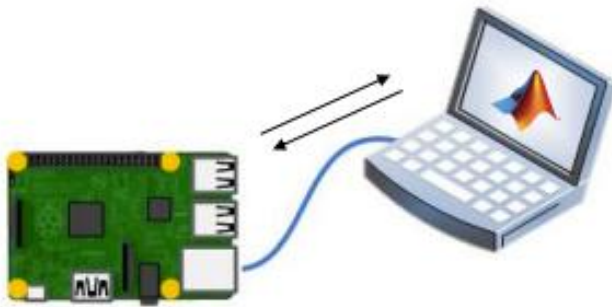
Example : Customer product intelligence



Lab : Arduino with Matlab and Simulink

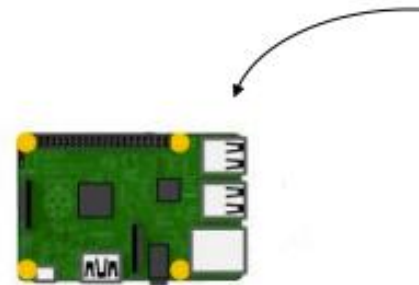
MATLAB Support Package

- Use hardware as I/O device
- Low-cost data acquisition
- Processing done on desktop
- Must be connected to computer

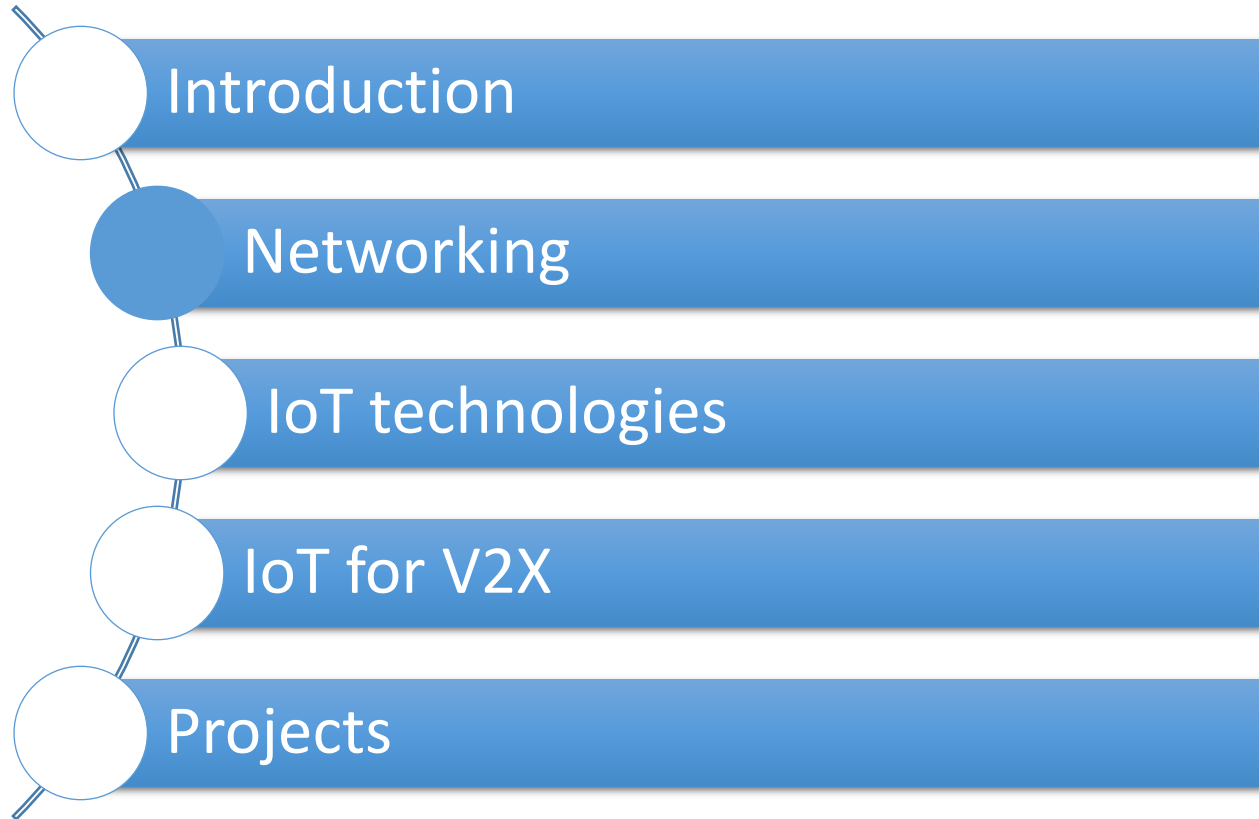


Simulink Support Package

- Develop algorithm in Simulink
- Deploy to hardware
- Processing done on hardware
- Can run independent of computer

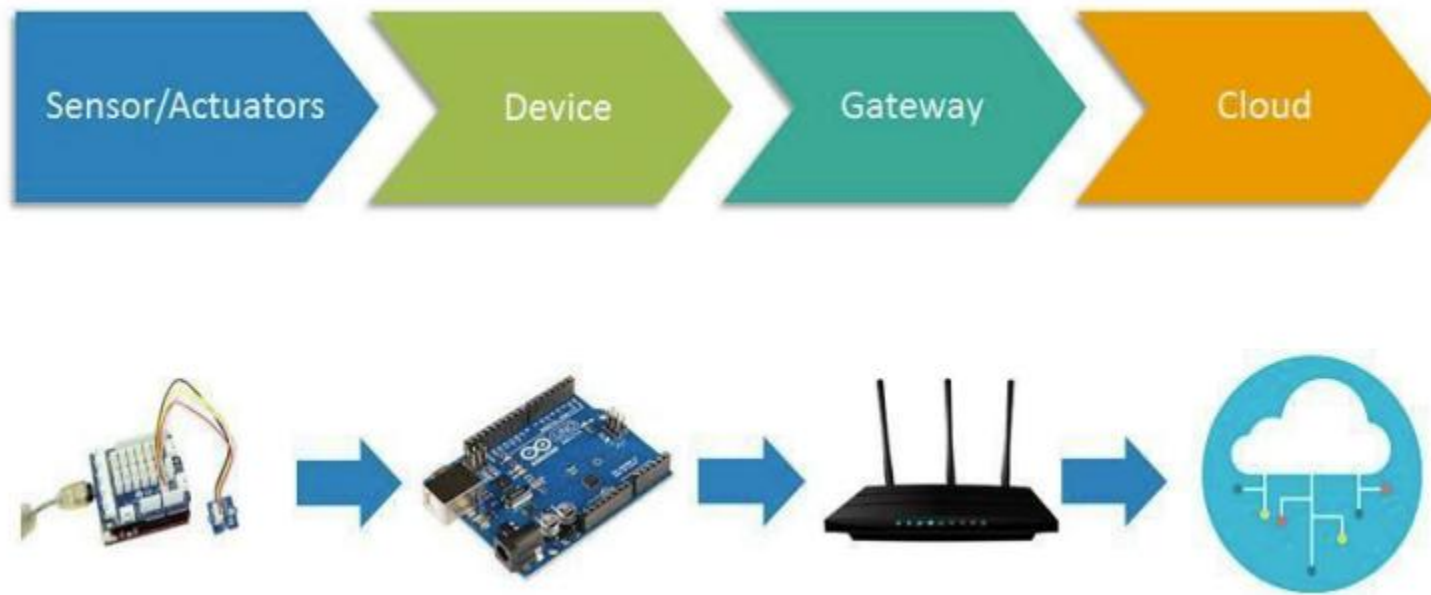


Agenda



IoT architecture

IoT architecture varies from solution to solution, based on the type of solution which we intend to build. IoT as a technology majorly consists of four main components, over which an architecture is framed



Networking

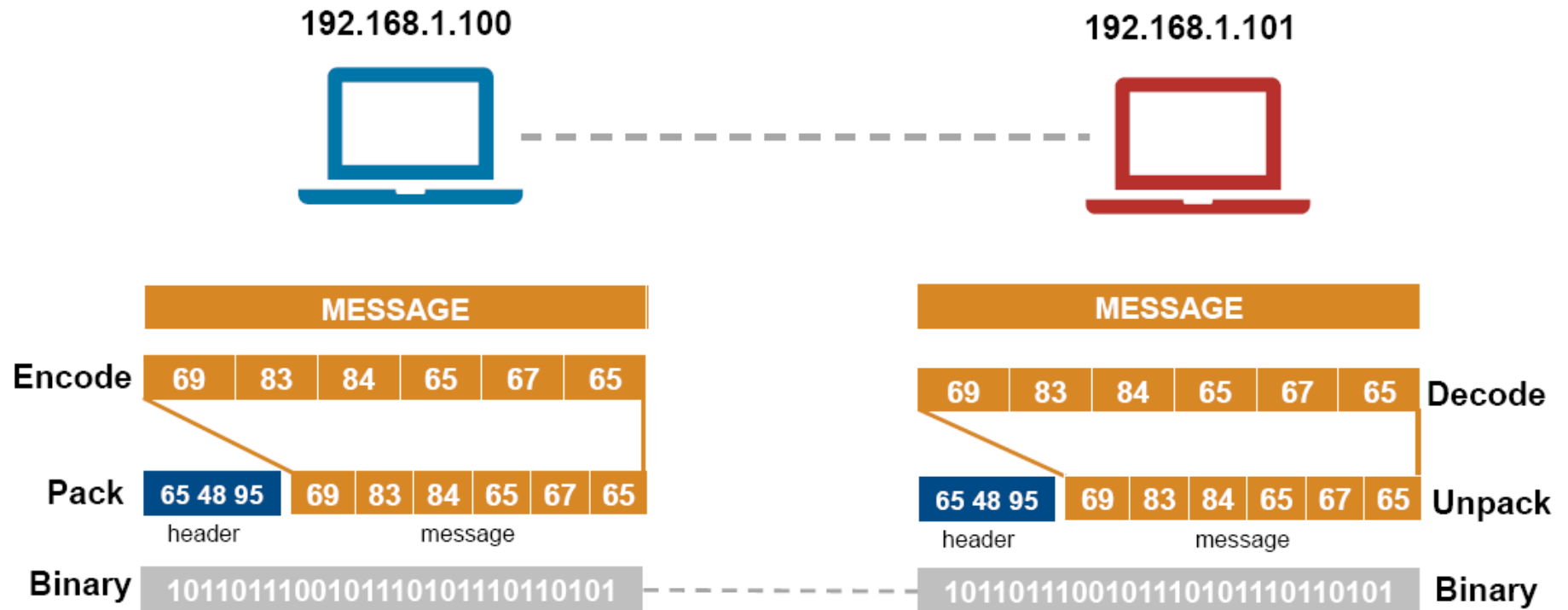
- Networking refers to the linking of computers and communication network devices (referred to as hosts), which interconnect through a network (Internet or Intranet) and are separated by unique device identifiers (Internet protocol, IP addresses and MAC addresses).
- These hosts may be connected by a single path or through multiple paths for sending and receiving data.
- The data transferred between the hosts may be text, images, or videos, which are in the form of binary.

How do systems communicate?



We need IP address and MAC address

Communication

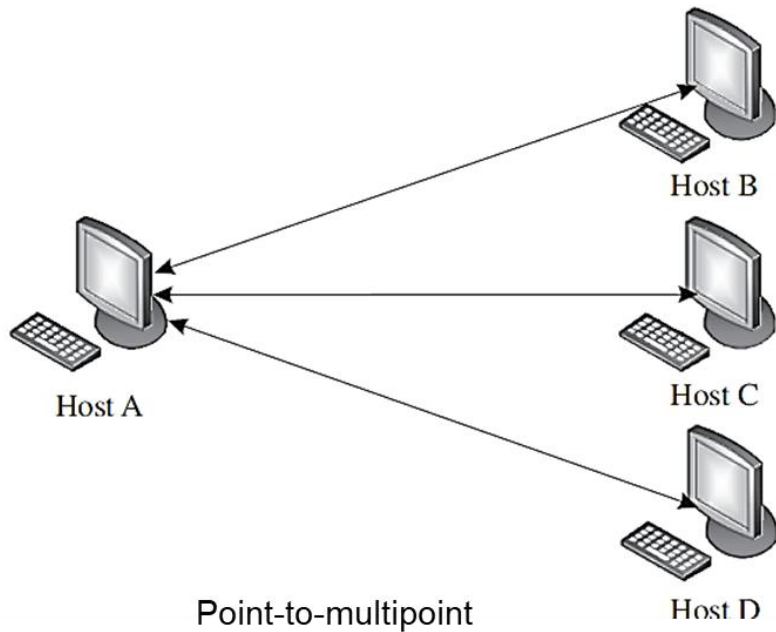
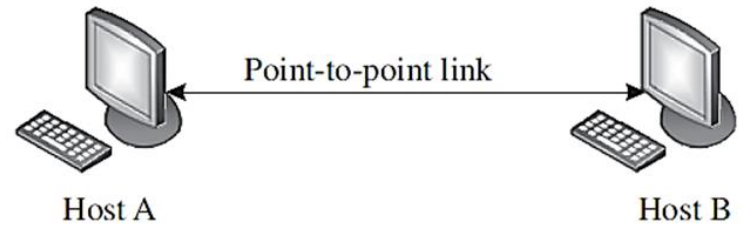


Networks classification

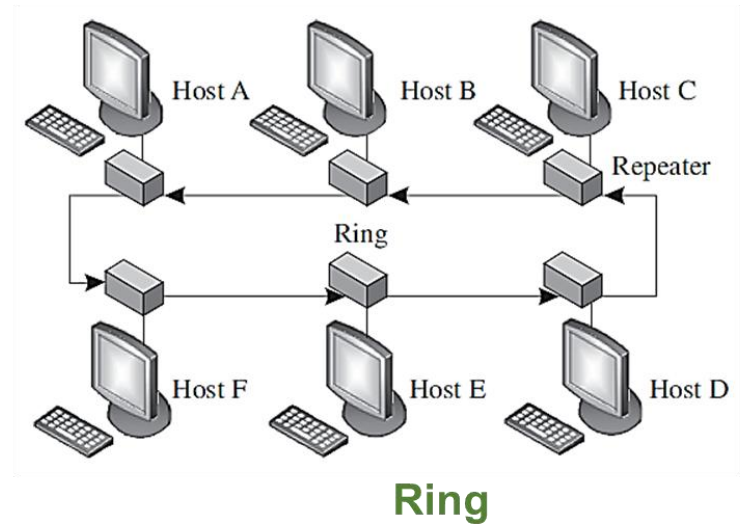
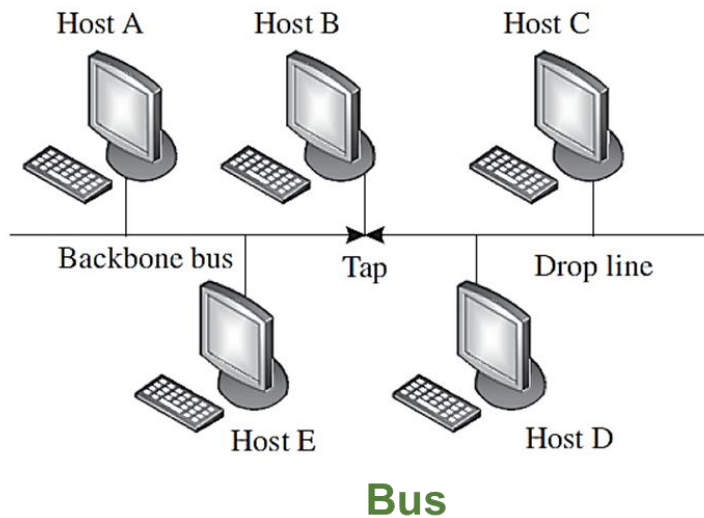
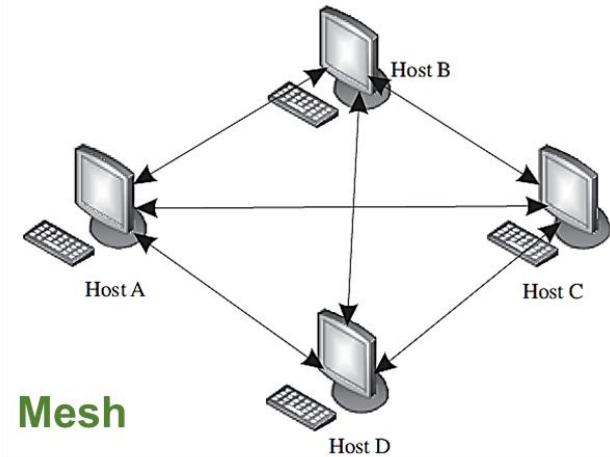
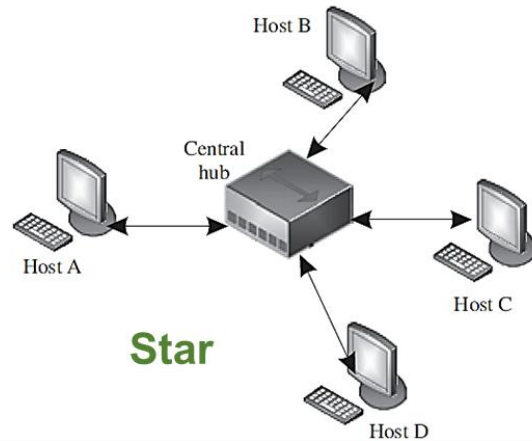
Computer networks are classified according to various parameters:

- 1) Type of connection
- 2) Physical topology
- 3) Reach of the network

Type of connection



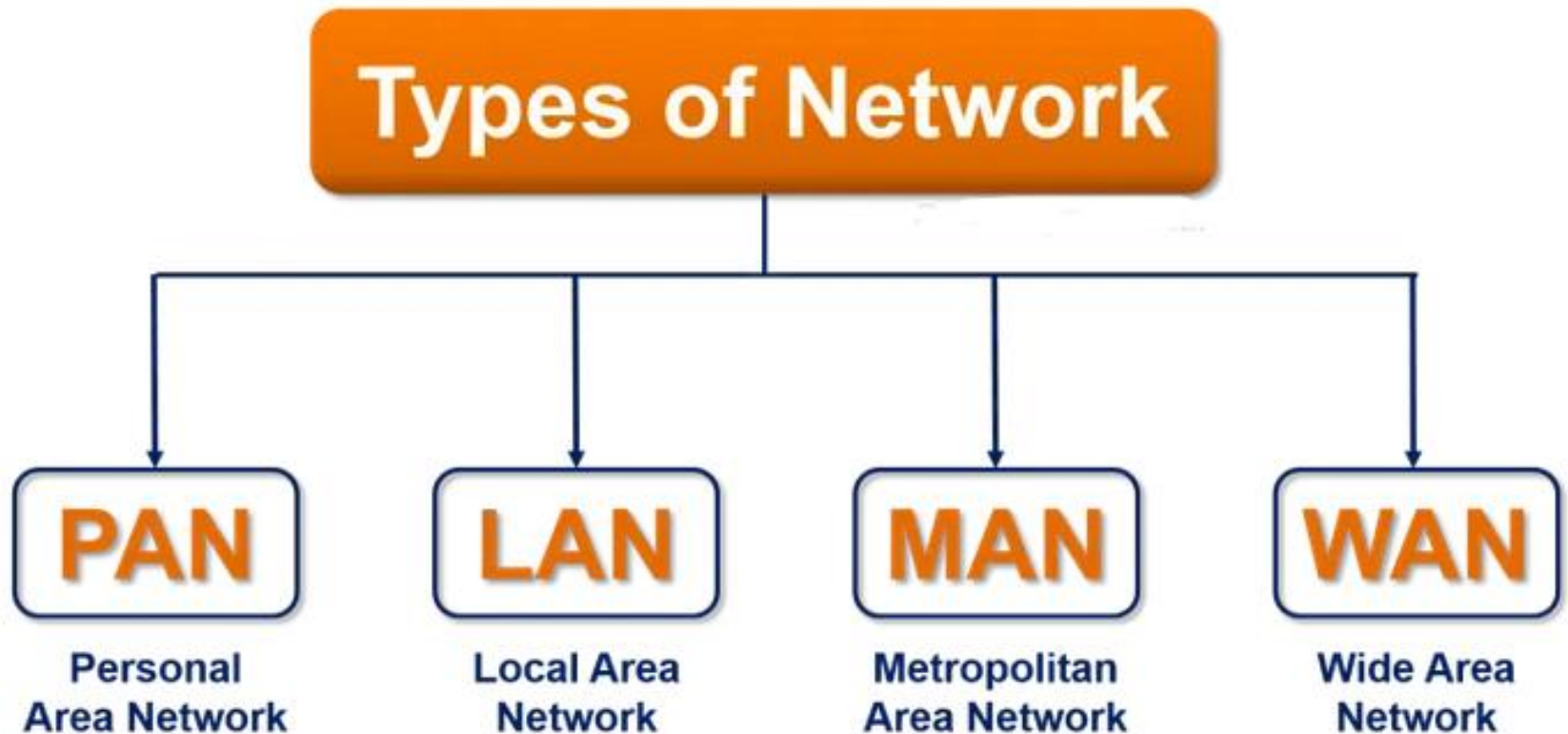
Physical topology



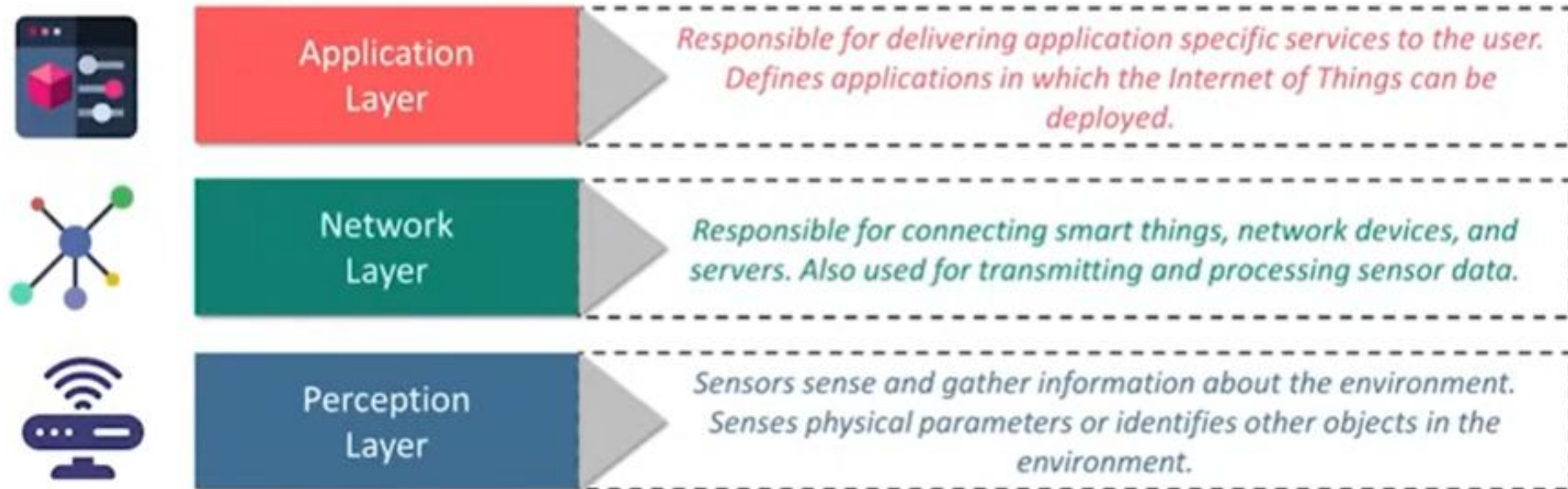
Topologies comparison

Topology	Feature	Advantage	Disadvantage
Star	Point-to-point	Cheap; ease of installation; ease of fault identification	Single point of failure; traffic visible to network entities
Mesh	Point-to-point	Resilient against single point of failures; scalable; traffic privacy and security ensured	Costly; complex connections
Bus	Point-to-multipoint	Ease of installation; cheap	Length of backbone cable limited; number of hosts limited; hard to localize faults
Ring	Point-to-point	Ease of installation; cheap; ease of fault identification	Prone to single point of failure

Network reachability



Basic architecture



Different Wireless communication technologies

Wireless Communication Technologies

Infrastructure-based

Infrastructureless

Broadcast

Cellular

Short Range

Medium Range

FM Radio,
DAB/DVB,
...

GSM
2G Cellular

UMTS
3G
Cellular

LTE /
WiMAX 4G
Cell.

LTE-V2X /
5G NR V2X

Millimeter,
Infrared,
Visible

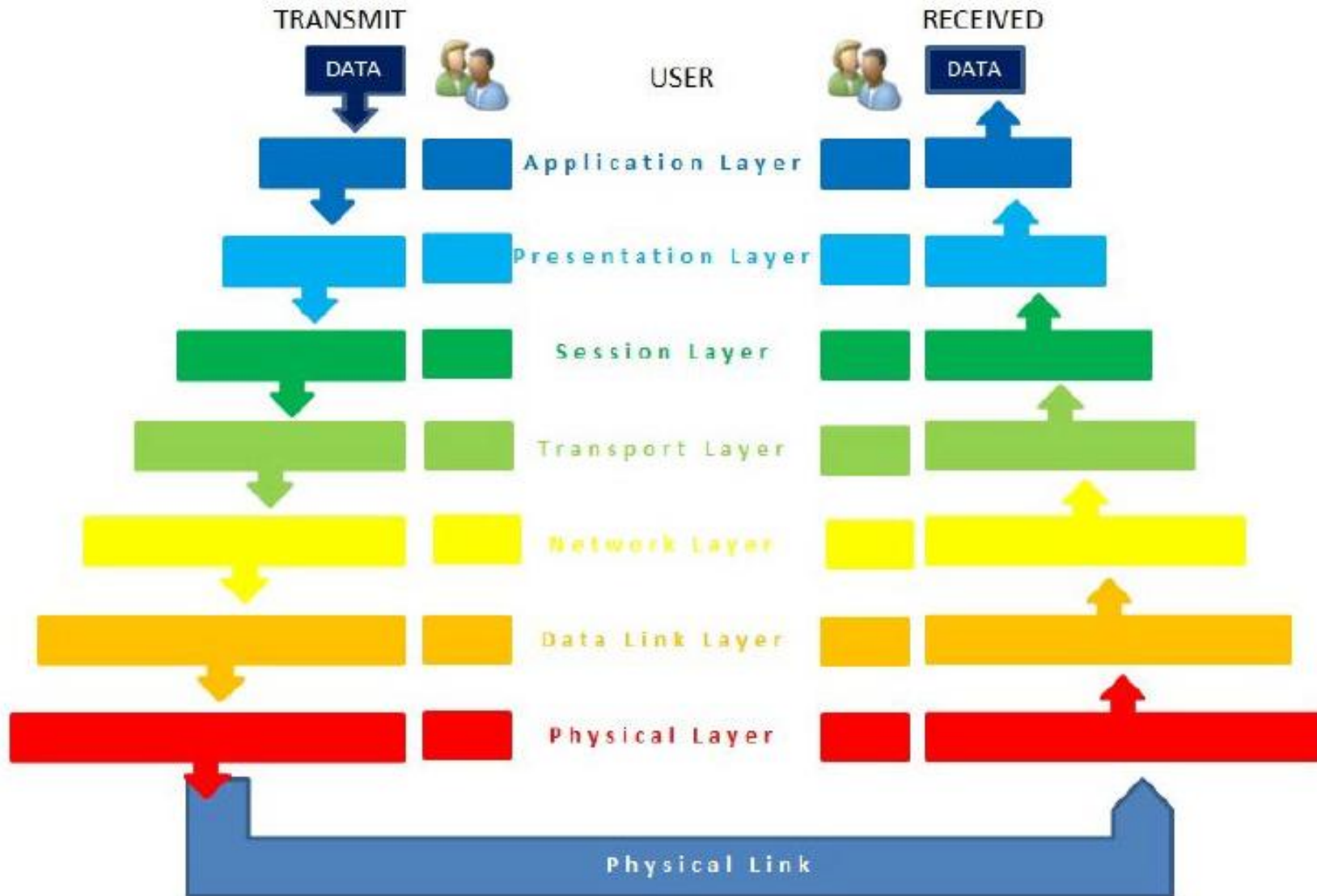
802.15.1
Bluetooth

802.15.4
ZigBee

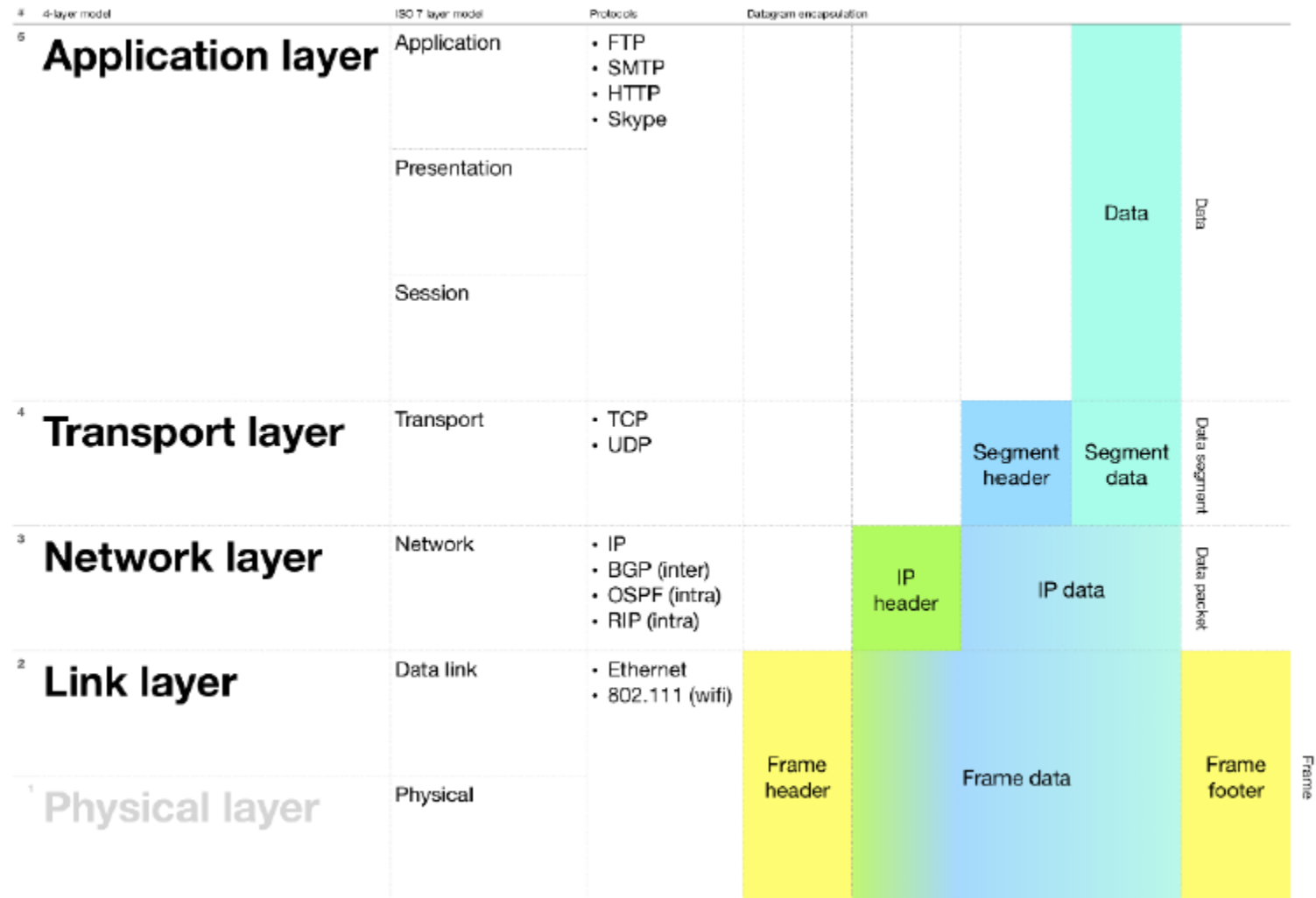
802.11p/bd
Wi-Fi

DSRC /
WAVE

Open System Interconnections (OSI) Layers

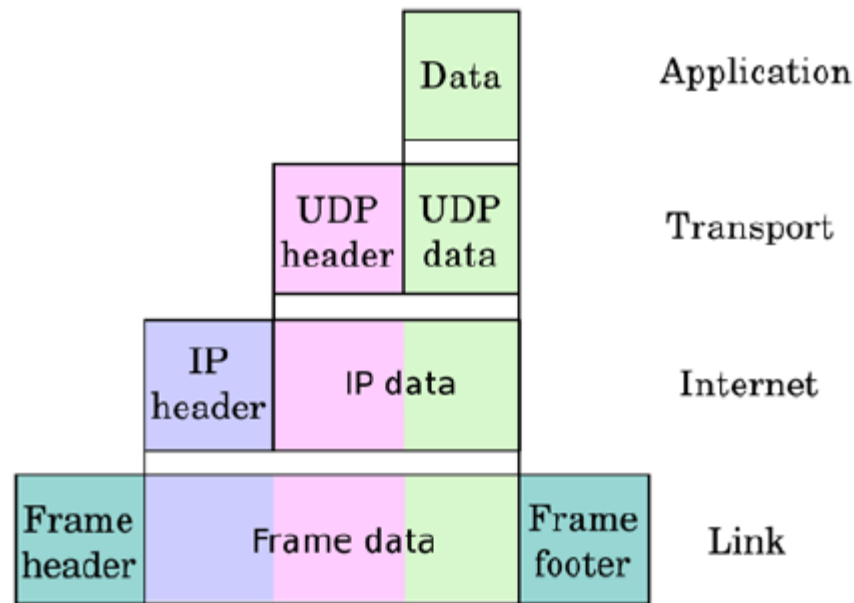


Protocols of the OSI model



Transport layer : UDP

User Datagram Protocol (UDP) is a very popular protocol for sensors such as lidars, cameras, and radars. It is a connection-less protocol. Connection-less, in this sense, just means that the protocol doesn't verify that it can reach the destination before it sends data. UDP lives on the transport layer of the OSI model.



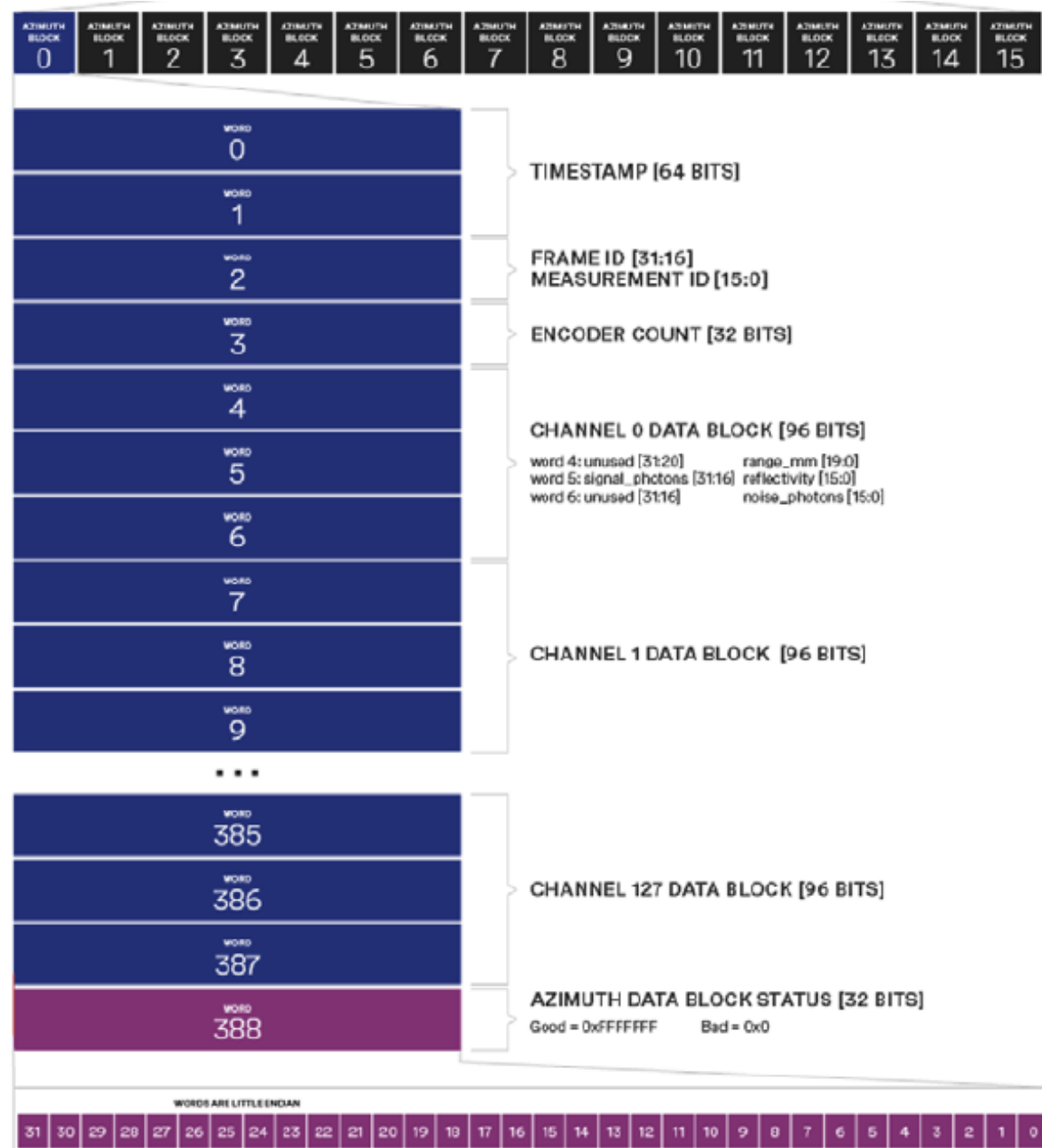
Transport layer : UDP protocol

UDP stores the source port, the destination port, the length of the data including the header, and finally a checksum. The checksum is simply a number that is created with an algorithm before the data is sent to ensure that when it is received, the data is intact and not corrupted. This is done by running the same algorithm on the received data and comparing the number generated to the checksum value.

Source Port Number(16 bits)	Destination Port Number(16 bits)
Length(UDP Header + Data)16 bits	UDP Checksum(16 bits)
Application Data (Message)	

UDP example

The UDP header is always 8 bytes (64 bits), while the data (message) can be up to 65,507 bytes in length. The following figure is a relevant example, to self-driving cars, of the data (message) field size of a UDP packet from a popular family of high-resolution lidar sensors:



Transport layer : UDP use

UDP is often used for **streaming-type** devices such as lidar sensors, cameras, and radars since it does not make sense to resend data if it is not received. Imagine you didn't receive a few azimuths in the lidar sensor example. Would it be useful to you to have that data resent to you? Probably not, since whatever the lasers bounced off is now in the past and likely in a different position. Another reason to use UDP for such devices is that due to the high data rates, it would slow things down tremendously to have to resend lost or corrupted data.

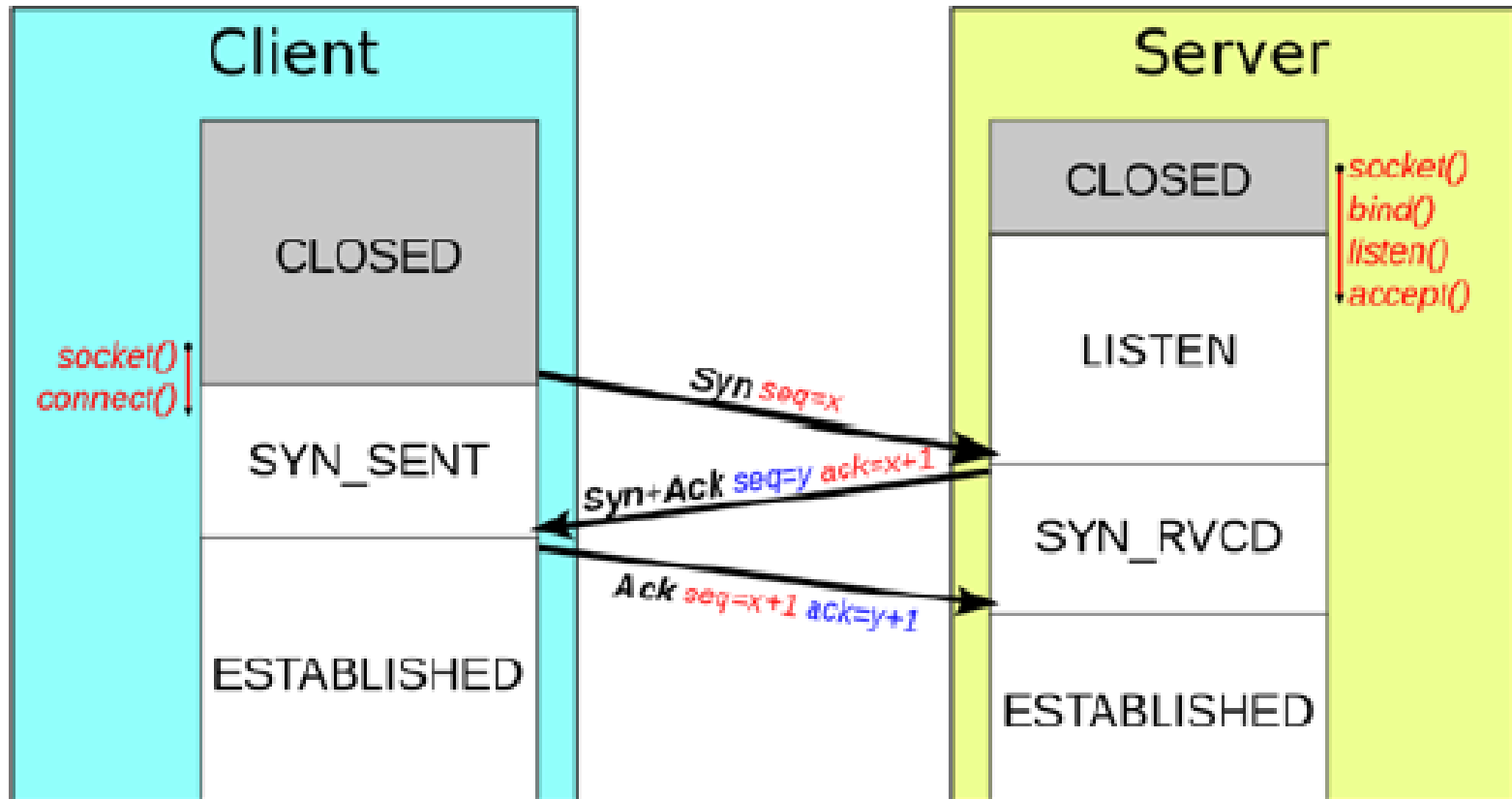
Transport layer : TCP

TCP is a connection-based protocol. This means that each time you want to send data, you need to do a three-way handshake. This is done through a process known as SYN-SYN/ACK-ACK.

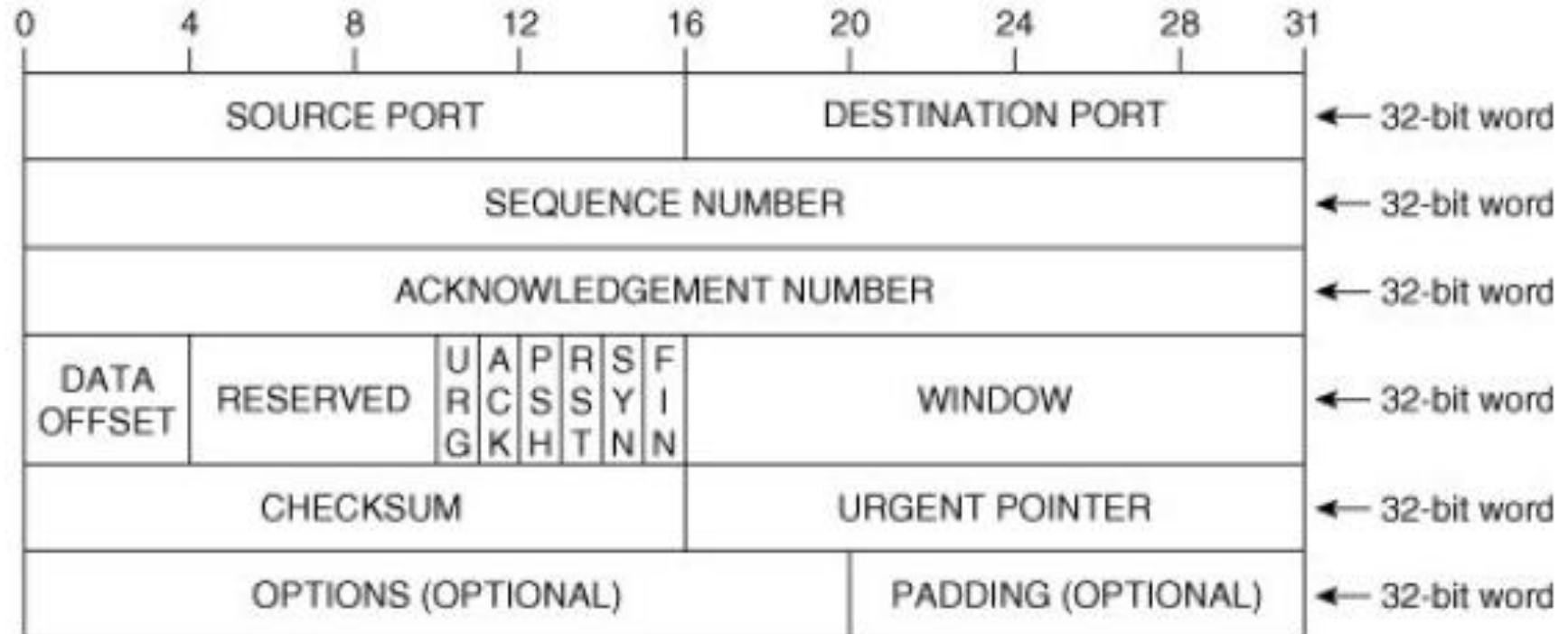
Transport layer : TCP connection verification

- SYN – The client sends a SYN (synchronization) packet with a randomly selected initial sequence number (x), which is used to count the bytes that are being sent. It also sets the SYN bit flag to 1 (more on this later).
- SYN/ACK – The server receives the SYN packet:
 1. It increments x by one. This becomes the acknowledgement (ACK) number ($x+1$), which is the number of the next byte it expects.
 2. It then sends a SYN/ACK packet back to the client with the ACK number as well as the server's own randomly selected sequence number (y).
- ACK – The client receives the SYN/ACK packet with the ACK number ($x+1$) and server sequence number (y):
 1. It increments the server's initial sequence number to $y+1$.
 2. It sends an ACK packet with the ACK number ($y+1$) and ACK bit flag set to 1 back to the server to establish the connection.

Transport layer : TCP connection diagram



Transport layer : TCP header



Network layer : IPv4

Every IPv4 address is made up of two pieces: a “network” portion, which tells routers what group of devices a packet should go to (e.g., a company, a campus, etc.) and a “host” portion which tells routers what specific device among that group the packet should go to.

How much of any given address is the network part and how much is the host part is determined by the “class” of the network. In each case, the part of the address not used for the network portion is left as the host portion.

Class	Network Portion	Hosts Allowed
A	from 1.0 to 127.0	approx. 16 million
B	from 128.0 to 191.255	65,536
C	from 192.0 to 223.255.255	255

Example : 198.41.12.151, 127.30.11.1

Network layer : Subnet mask

A subnet mask tells a router how much of an address it should treat as the network portion. The masks for traditional Class A, B and C networks are shown below.

Class	Subnet Mask
A	255.0.0.0
B	255.255.0.0
C	255.255.255.0

Comparing the masks above to the first chart, you can see that the 255s in a mask identify the network portion of the address

Network layer : Broadcast address

The broadcast address is the address to which devices send packets meant for all other devices. All devices “listen” for broadcasts in addition to their own address. Address Resolution Protocol (ARP) packets and routing information are examples of packets sent to the broadcast address. Most often, the broadcast address is the last address in the network (or subnet) with the host portion being all 1’s binary (some networks use 0.0.0.0 or 255.255.255.255, however). Table A-4 shows some examples of broadcast addresses.

Class	Network	Subnet Mask	Broadcast
A	45.0.0.0	255.0.0.0	45.255.255.255
B	128.138.0.0	255.255.0.0	128.138.255.255
C	198.41.9.0	255.255.255.0	198.41.9.255

Network layer : IPv6

Despite the success of IPv4, critics started arguing that IPv4 was insufficient for new applications, if we assign an IP address to each device (e.g., each smart appliance, each car, each smart watch), the address space will run out.

IPv6 (Internet Protocol version 6) offers several advantages over IPv4, addressing limitations and improving functionalities. Here are the main benefits of IPv6 over IPv4:

1. Larger Address Space:

IPv4: 32-bit addresses, supporting about 4.3 billion unique IP addresses.

IPv6: 128-bit addresses.

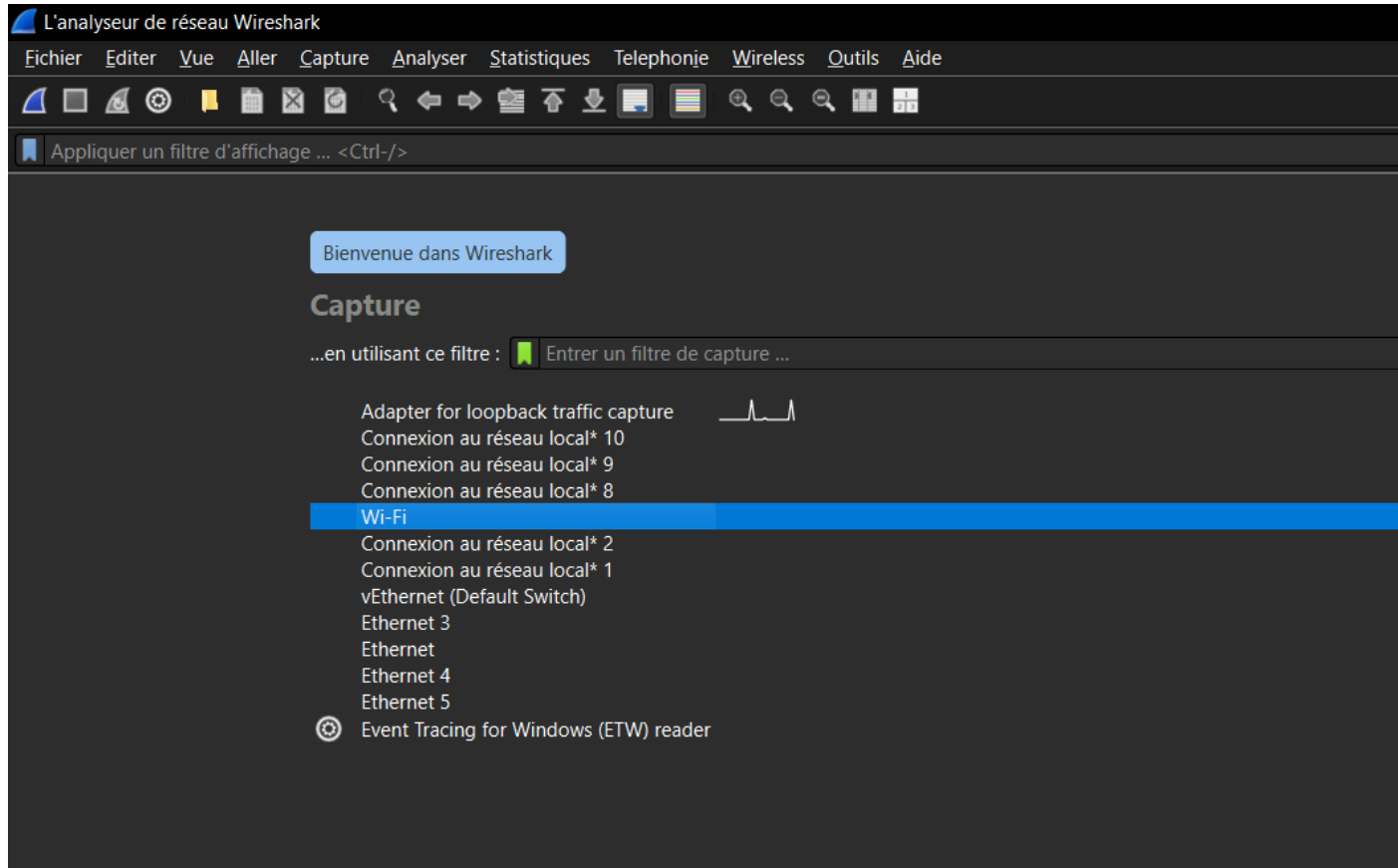
2. Simplified Header Format:

IPv6 has a more efficient and simplified header compared to IPv4. This improves packet processing speed and reduces the workload on network devices

3. Better Security:

IPv6 was designed with security in mind and supports IPsec (Internet Protocol Security).

Example : CMD et Wireshark



Lab : Arduino IoT communication

Exercise 1 :

Make an UDP communication between the Arduino IoT and the computer and send a constant value

Analyze the UDP header

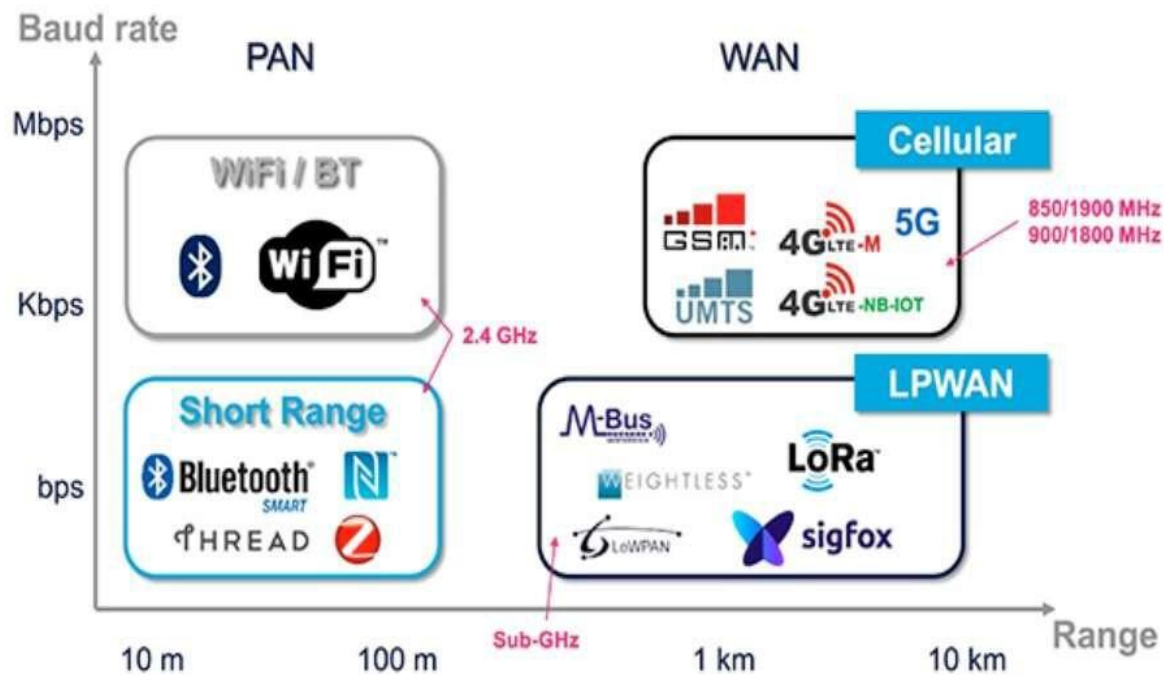
Exercise 2 :

Make an UDP broadcast of the accelemoter values

Read the signal in different Arduino cards

Analyze the UDP header

IoT technologies



Network layer : Zigbee

In addition to connecting conventional computers, an internet can be used by embedded devices. The concept of connecting devices is sometimes referred to as an **Internet of Things**.

IEEE has created standard **802.15.4** for a **low-power** wireless network technology intended to support connections of small embedded devices. The low-power aspect makes 802.15.4 radios attractive for devices that run on battery power. A consortium of vendors has chosen the term **ZigBee** to refer to products that use IEEE's 802.15.4 standard for radios and run a specific protocol stack that includes IPv6 plus protocols that allow a set of wireless nodes to organize themselves into a mesh that can forward packets to and from the Internet.

BLE Communication protocol

Standard BLE protocol structure:

Services: Logical containers for related data

Characteristics: The actual data points you read/write to

UUIDs: Unique identifiers for services and characteristics

Example : Phone connection

1. Discover services - you'll see your custom service

2. Write a number:

Tap the write characteristic (UUID ending with 9ABC)

Choose format: UTF-8 or Byte

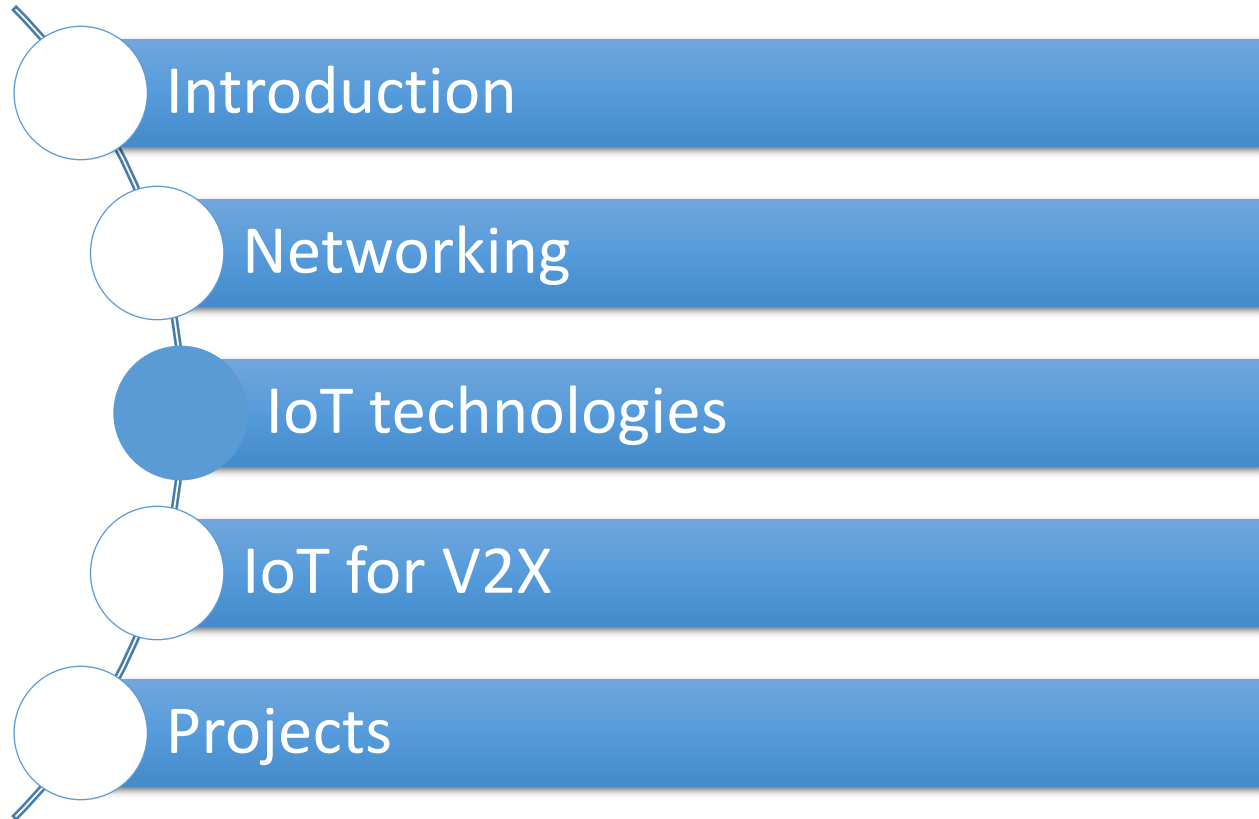
Enter: 42 (this sends the number 42)

Lab : Arduino IoT communication

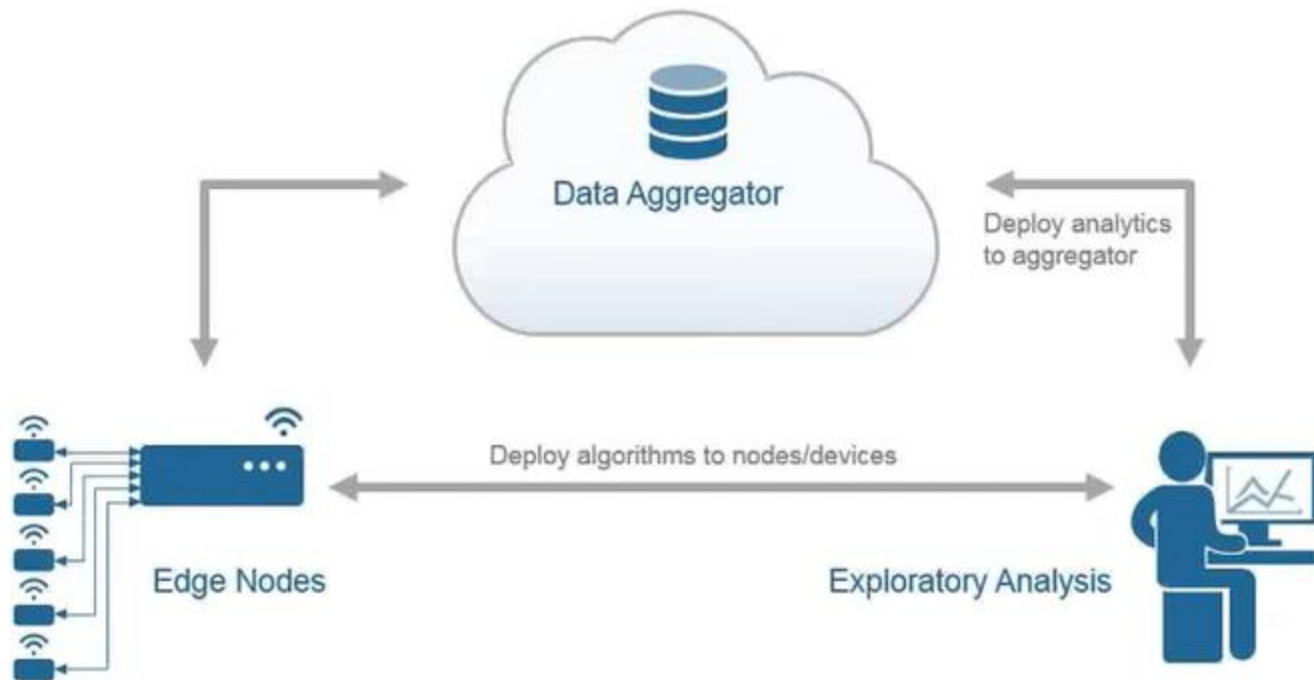
Exercise 3 :

Bluetooth communication with nrfConnect App

Agenda



IoT Process



Thingspeak

- Free online data aggregation platform
 - Typically used to collect data from sensors ("Things")
 - Provides instant visualization of the data
 - Popular for people experimenting in IoT
- Can be used to act on data
 - E.g. Tweet a message when the temperature in your backyard reaches 32 degrees
- Can be used to analyze data
 - MATLAB integration allows users to run scheduled MATLAB code on data coming into ThingSpeak



 **Collect**
Send sensor data to the cloud.

 **Analyze**
Analyze and visualize your data.

 **Act**
Trigger a reaction.

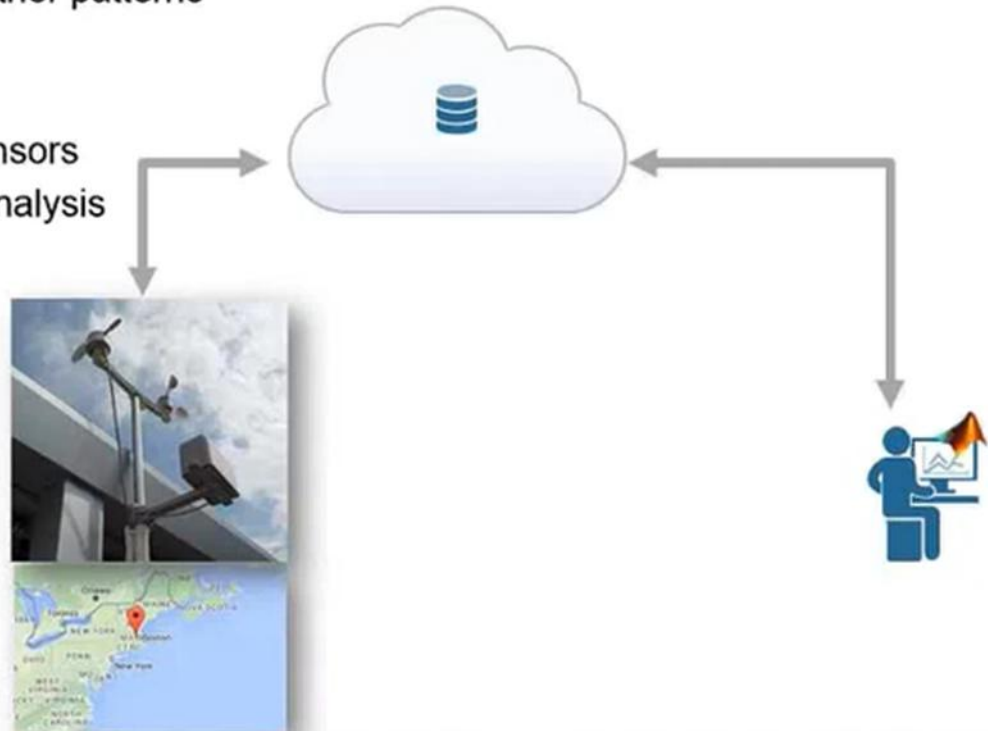
Example1 : weather station

Objectives

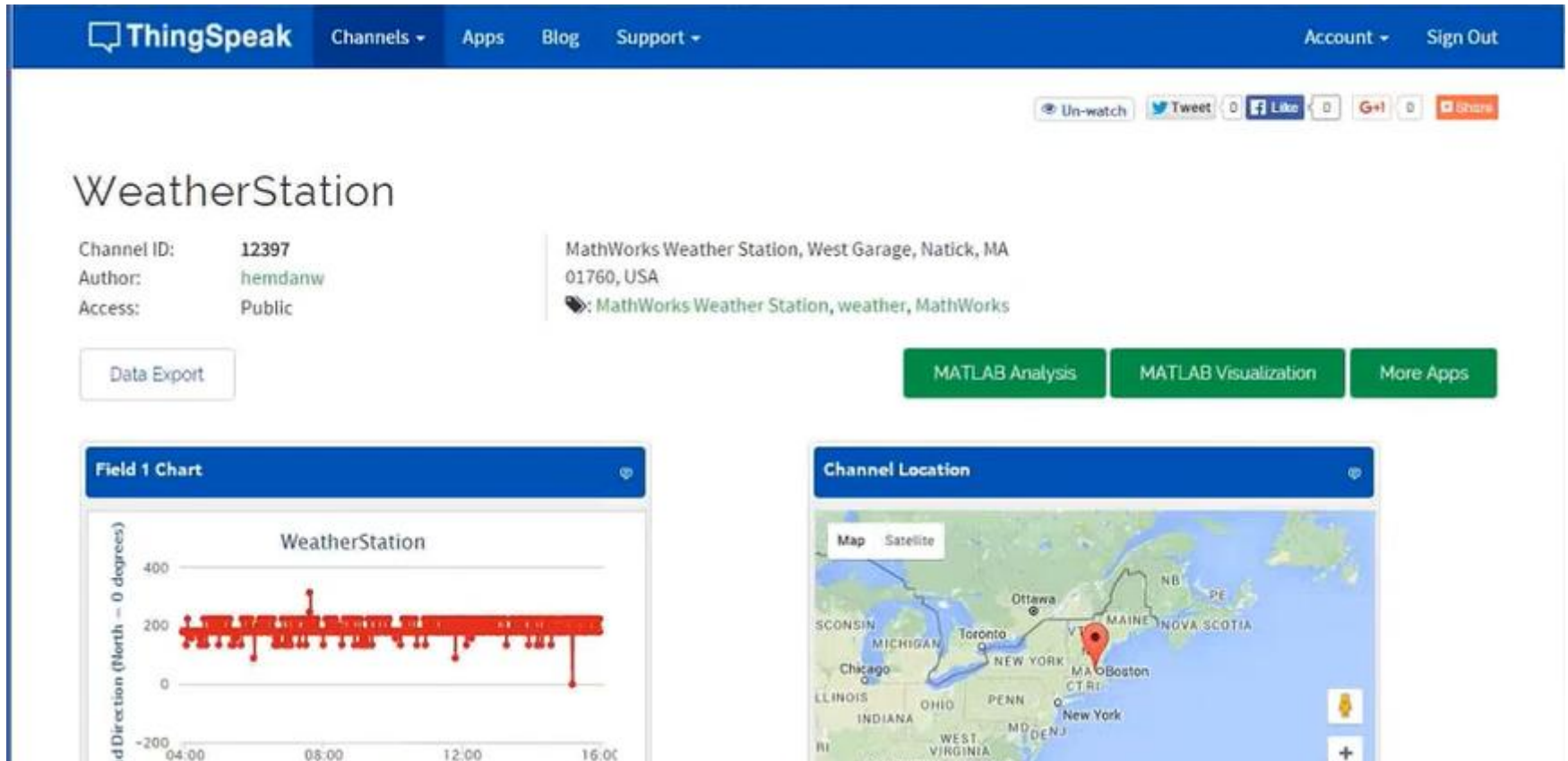
- Measure, explore, discover weather patterns

Solution

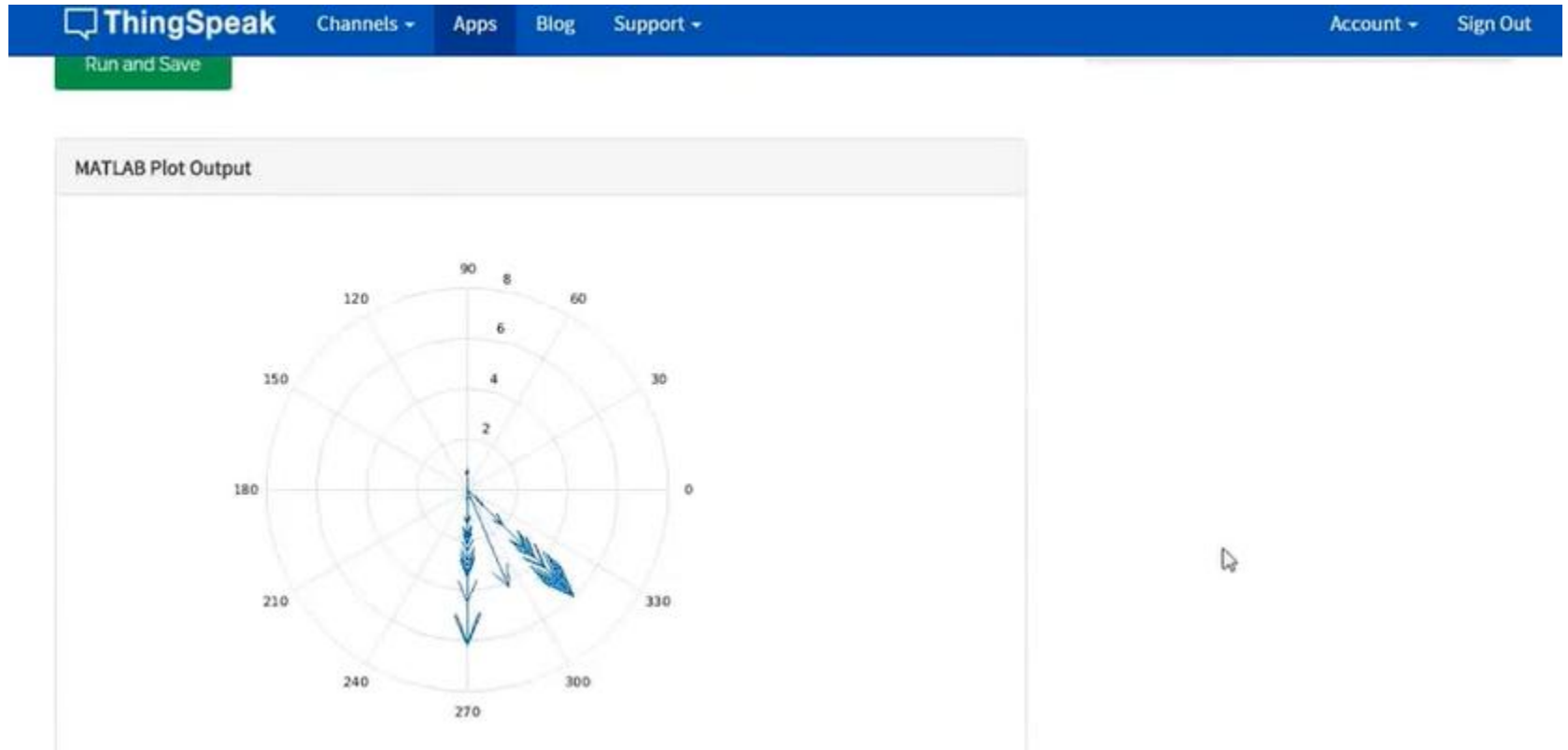
- Arduino station with weather sensors
- Cloud-based aggregation and analysis



Example1 : weather station



Example 1 : weather station



Example 1 : weather station

 Channels ▾ Apps Blog Support ▾ Account ▾ Sign Out

Apps / MATLAB Analysis / Calculate Dew point Now

Name
Calculate Dew point Now

MATLAB Code

```
36 % Convert temperature from Fahrenheit to Celsius
37 tempC = (5/9)*(tempF-32);
38
39 % Calculate dew point
40
41 % Specify the constants for water vapor (b) and barometric (c) pressure.
42 b = 17.62;
43 c = 243.5;
44 % Calculate the intermediate value 'gamma'
45 gamma = log(humidity/100) + b*tempC ./ (c+tempC);
46 % Calculate dew point in Celsius
47 dewPoint = c*gamma ./ (b-gamma);
48 % Convert to dew point in Fahrenheit
49 dewPointF = (dewPoint*1.8) + 32;
50
51 display(dewPointF, 'Dew point')
52
53 % Write the dew point value to another channel specified by the
```

Help
My Channels Documentation
New Channel

Channel Info
Name: Car Counter
Channel ID: 38629
Access: Public
Write API Key: R3JR5IB50XTHI9QX
Fields:
1: Number of Westbound Cars
2: Number of Eastbound Cars
Name: Daily Traffic Volume
Channel ID: 51671
Access: Public
Write API Key: 7MTIDH1RPW69B2XE
Fields:

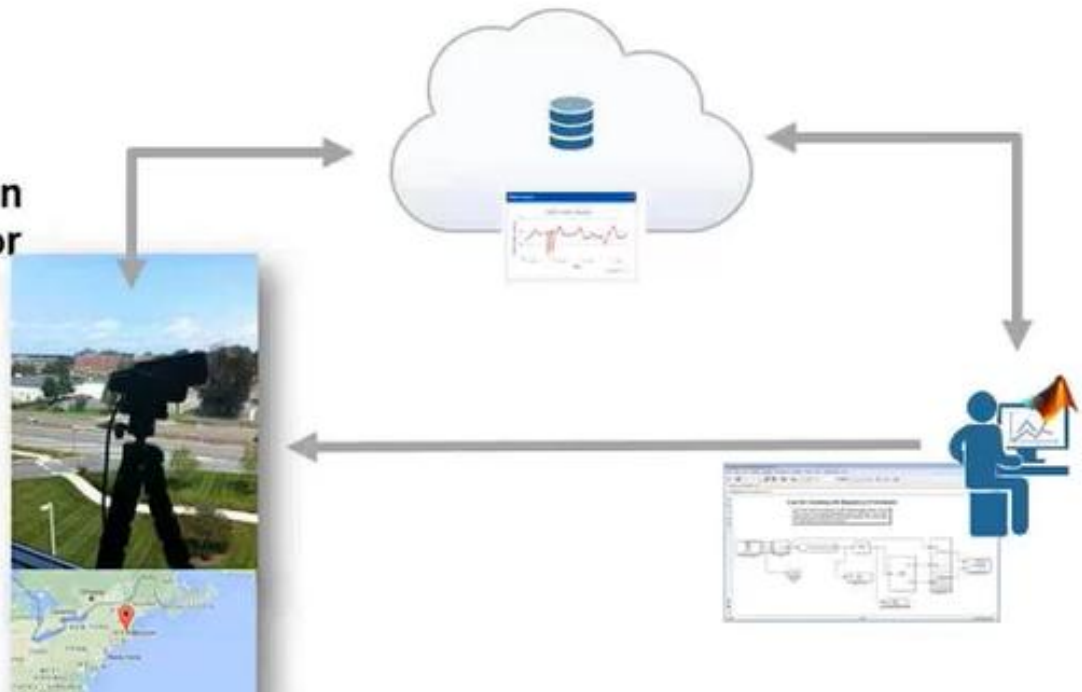
Example 2 : Car counter

Objectives

- Measure, explore, discover traffic patterns

Solution

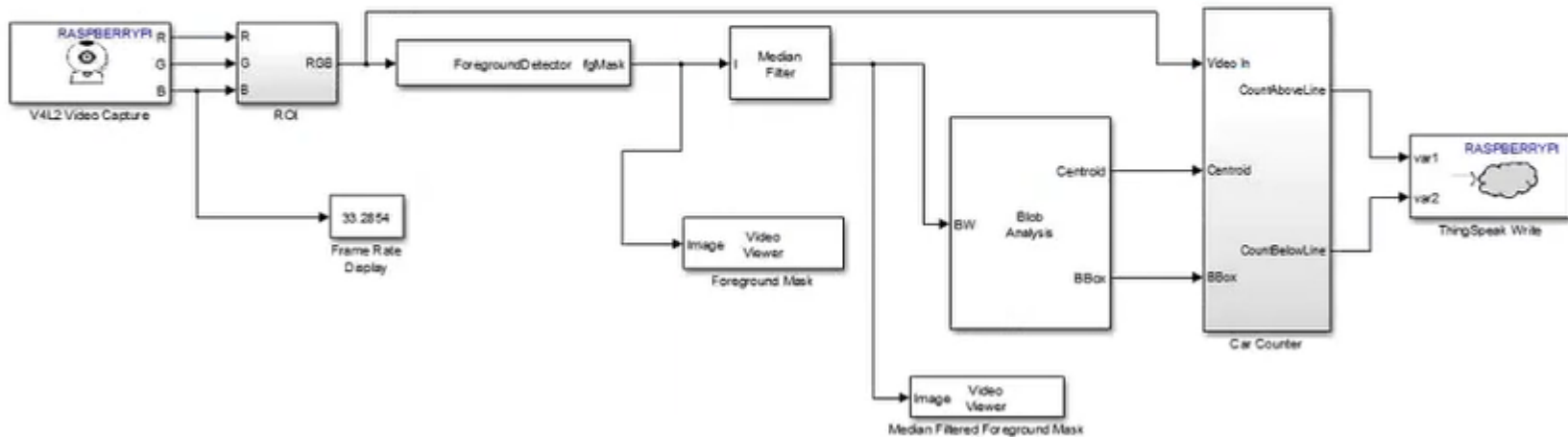
- RaspberryPi 2+ webcam
- **Automated deployment of vision algorithms on embedded sensor**



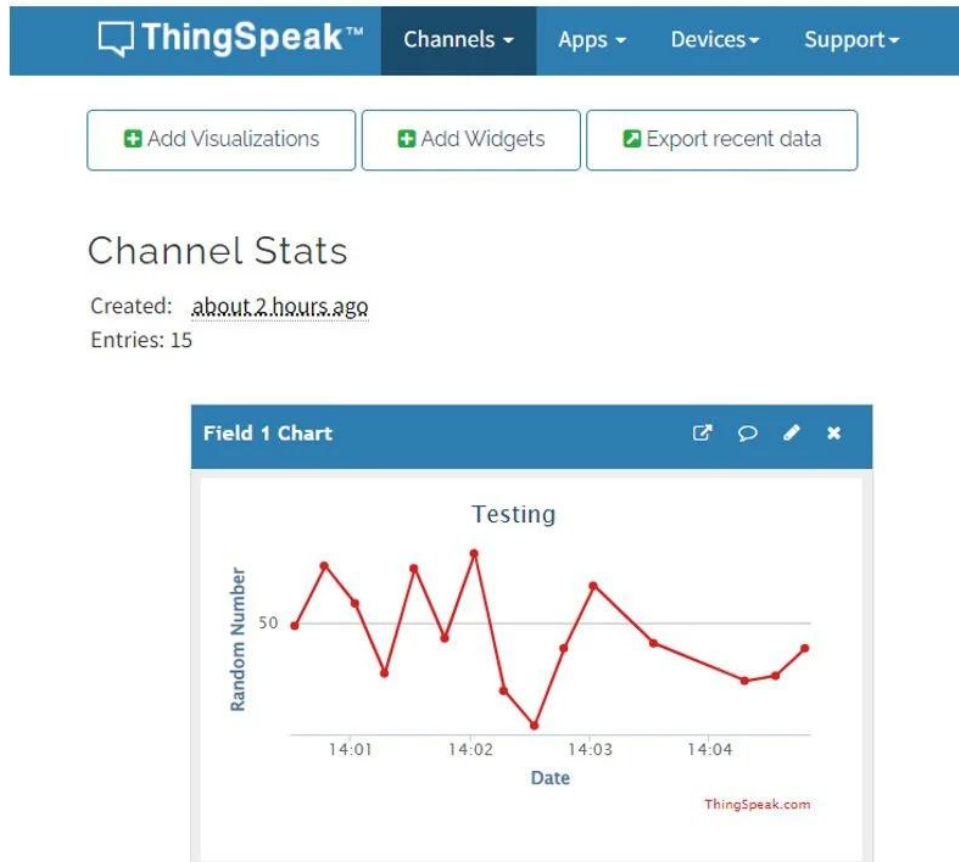
Example 2 : Car counter

Live Car Counting with Raspberry Pi Hardware

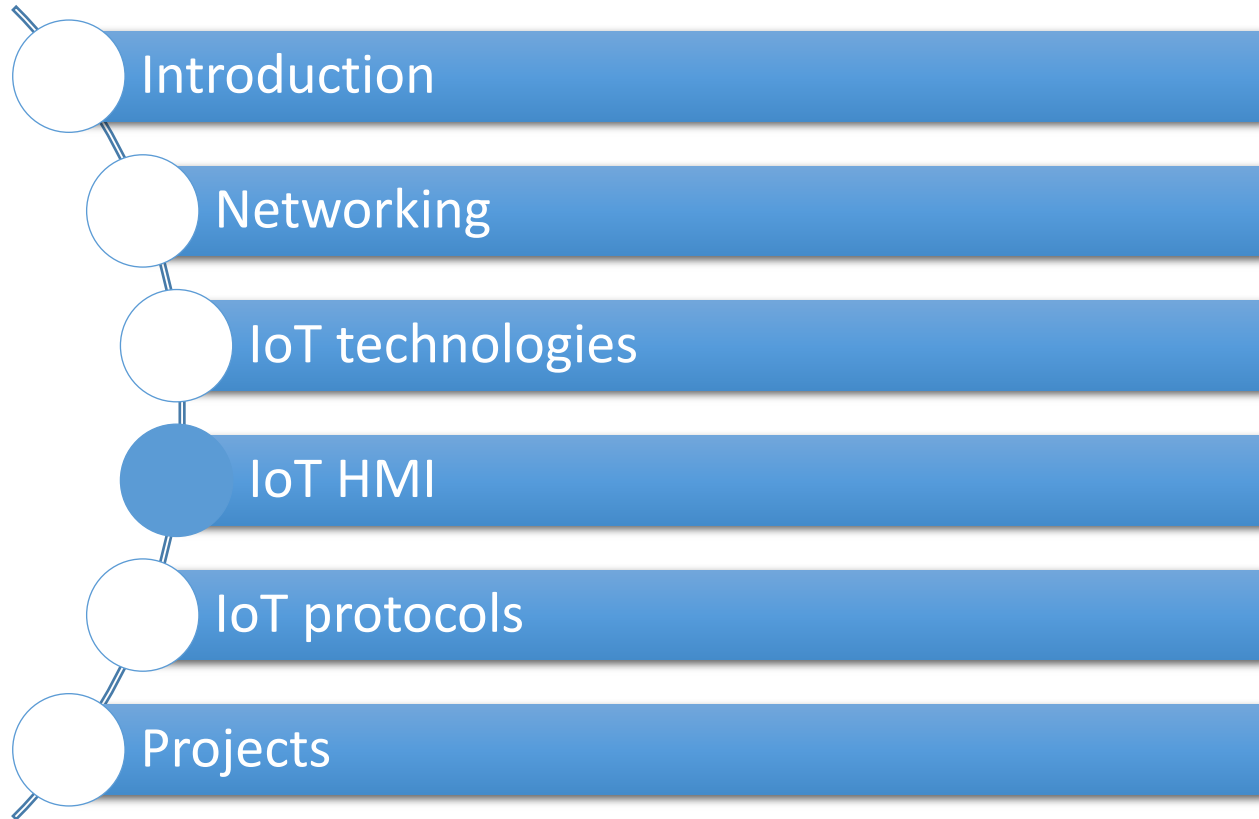
To run this model on hardware, on the Simulink Editor toolbar, click the "Run" button. Note that the simulation mode is "External". This mode allows you to tune parameters and monitor signals in the model while the application is running on hardware.



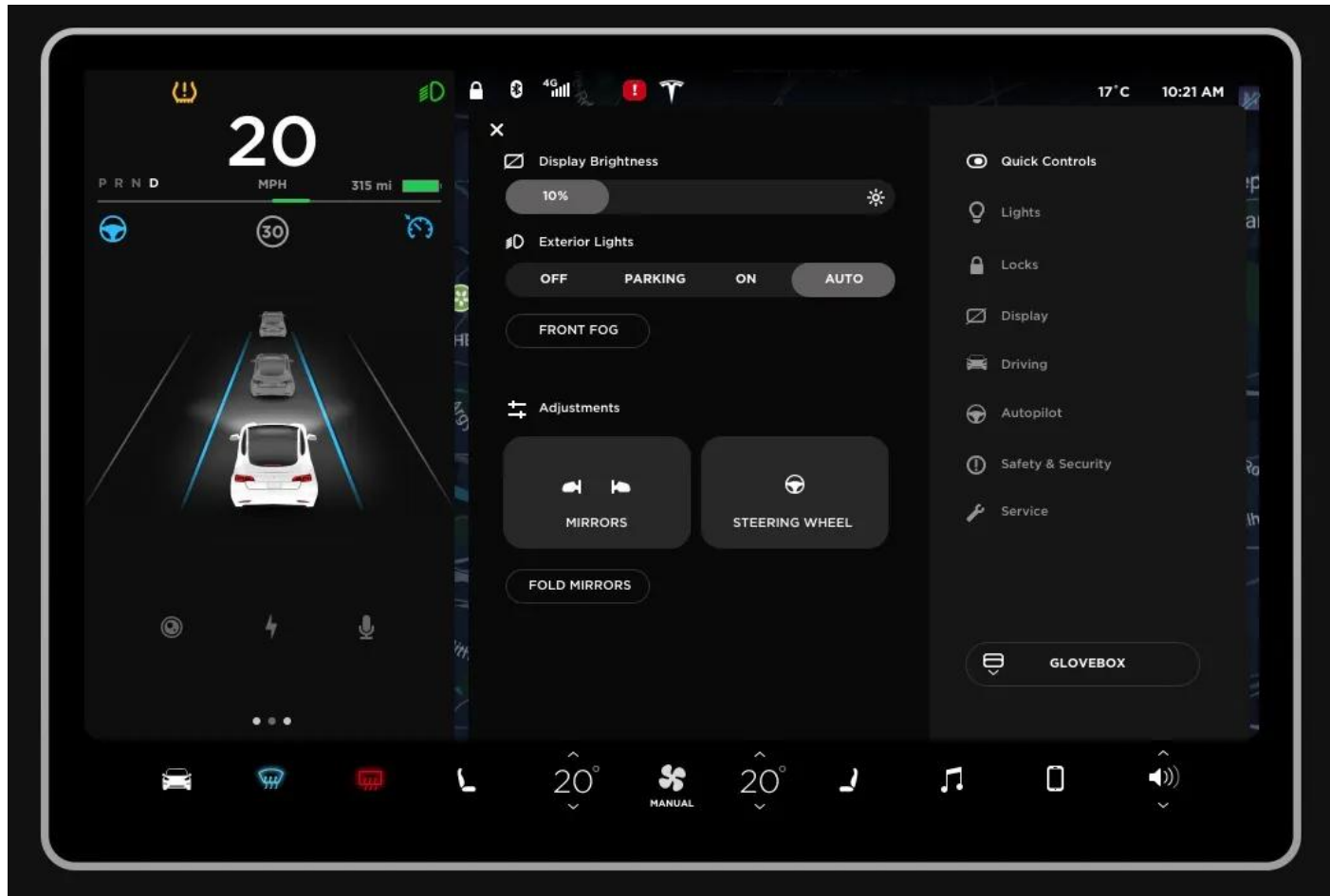
Lab : Send sensor data to the cloud for analysis



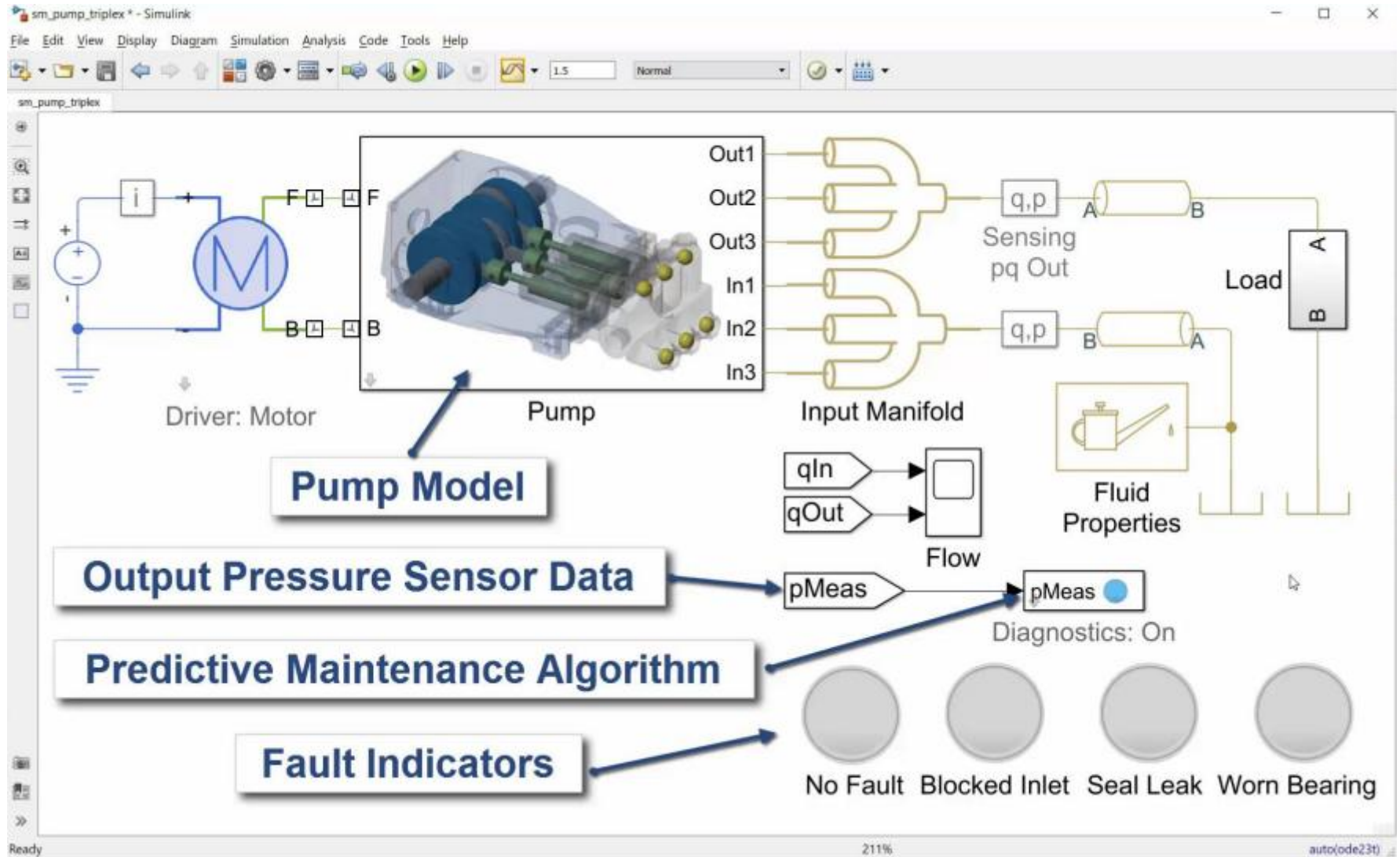
Agenda



HMI in automotive industry



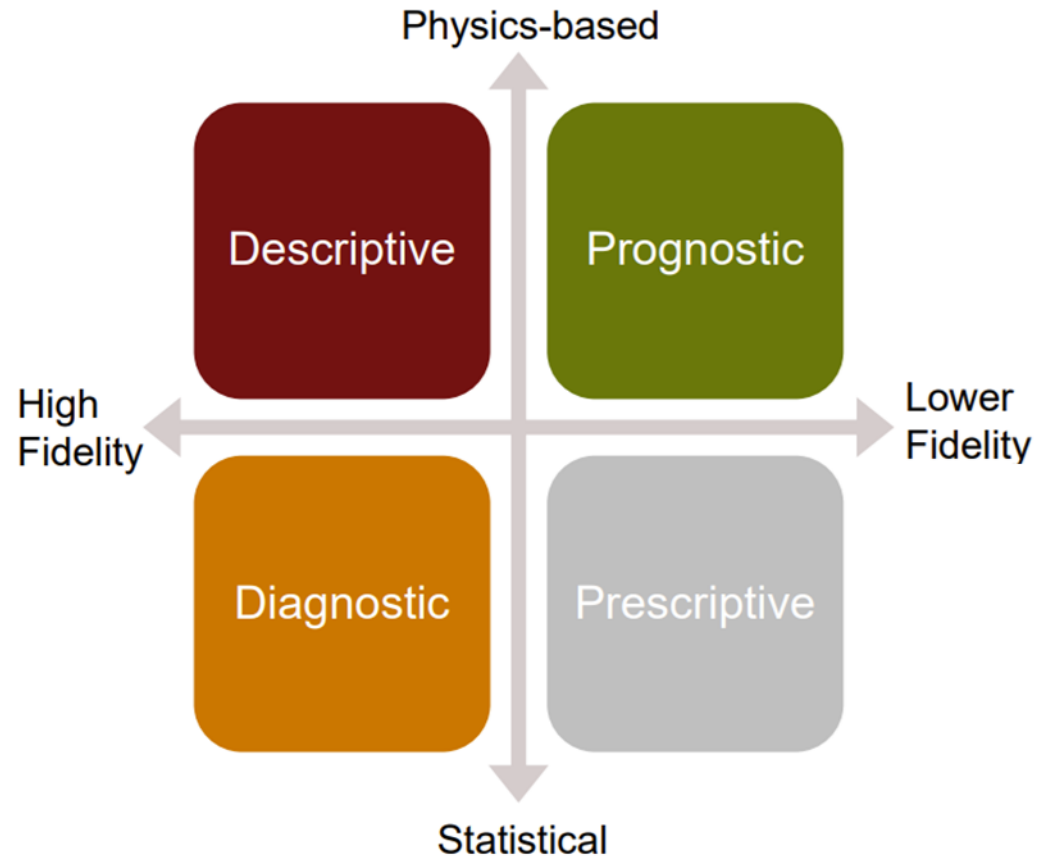
Simulink HMI components



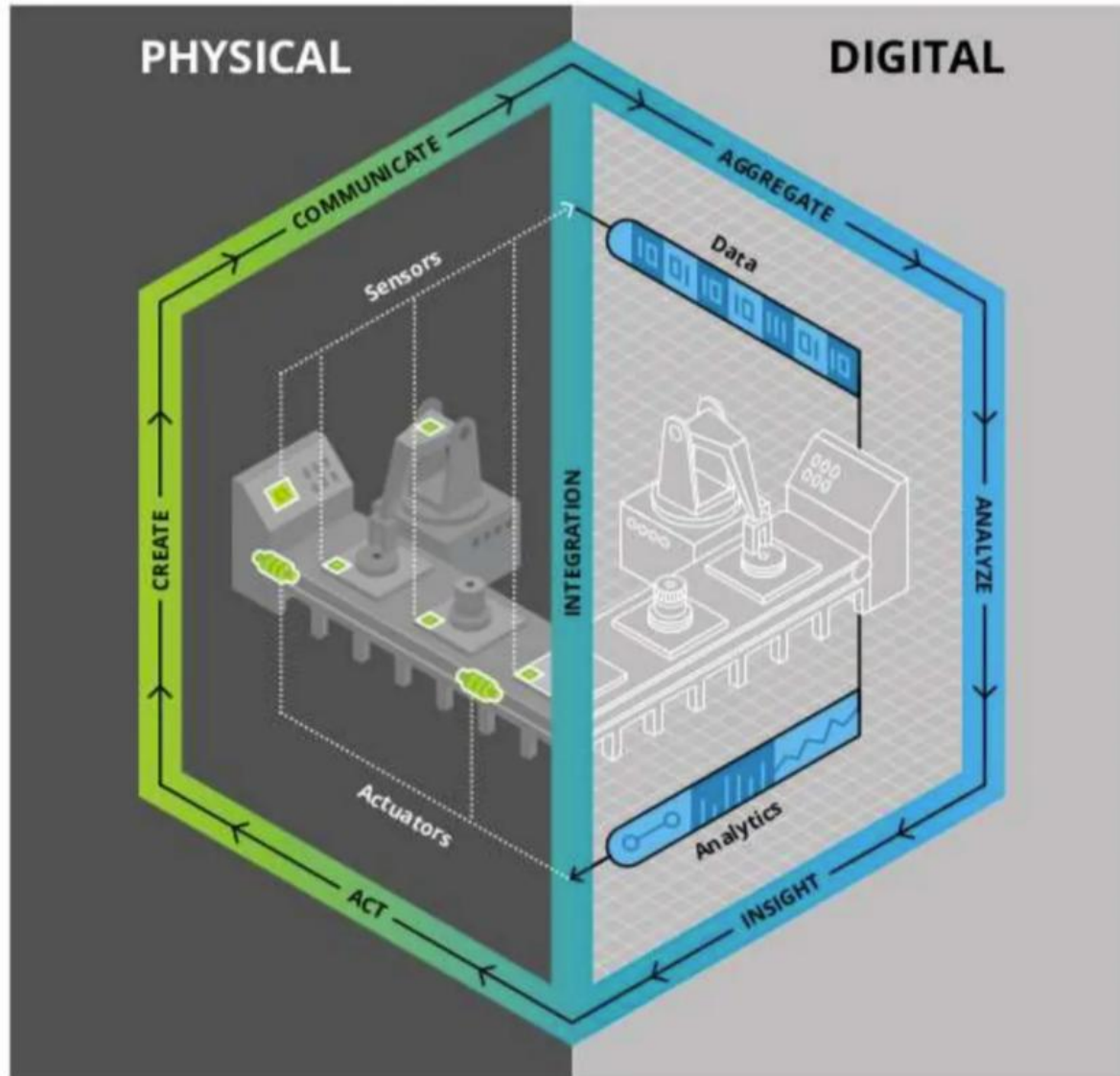
Digital Twin

Digital twins refer to computerized companions of physical assets that can be used for **various purposes**.

- **Digital twins** use data from sensors installed on physical objects to represent their near real-time status, working condition or position.
- **Digital twins** can be also used for monitoring, diagnostics and prognostics.



Digital Twin

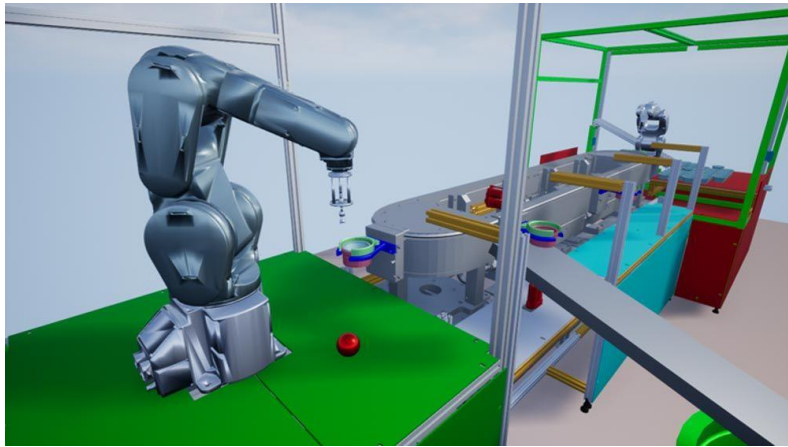


3D Animation

3D Animation provides an interface to a simulation environment that is visualized using the Unreal Engine® from Epic Games®. Within this environment, you can create a 3D environment with actors. An actor is any object in the 3D environment that can move and rotate to support 3D transformations.

In MATLAB, use the **sim3d.World** object and functions to create and view a 3D environment. The **sim3d.Actor** object and functions create actor objects and builds appearance for actor in 3D environment.

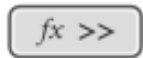
Examples



Lab : Simulink Dashboard and Digital Twin

▼ Dashboard

- ▶ Automotive Indicator Lamps
- ▶ Basic Shape Icons
- ▶ Customizable Blocks
- ▶ Wireless Icons



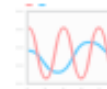
Callback Button



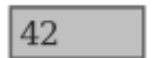
Check Box



Combo Box



Dashboard Scope



Display



Edit



Gauge



Half Gauge



Knob



Lamp



Linear Gauge



MultiStateImage



Push Button



Quarter Gauge



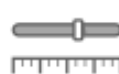
Radio Button



Rocker Switch



Rotary Switch



Slider

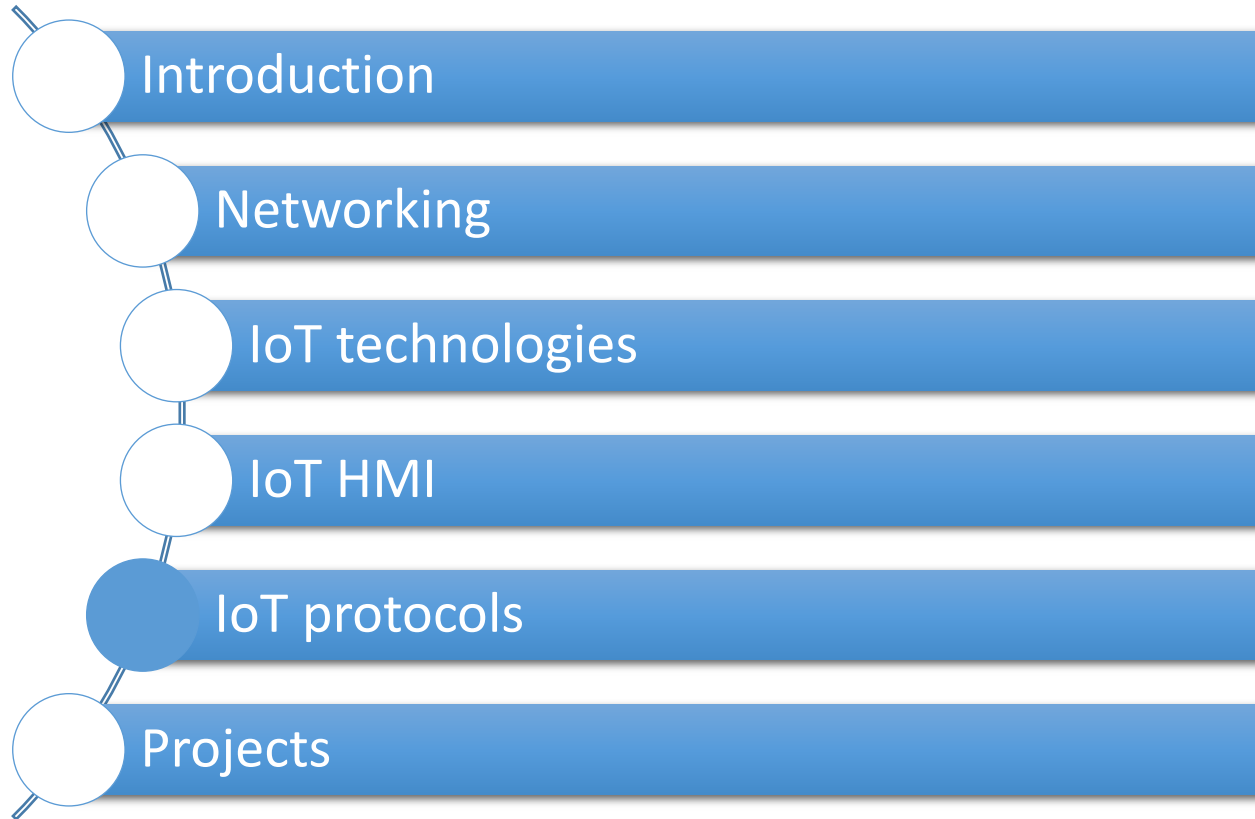


Slider Switch



Toggle Switch

Agenda



Websocket

WebSocket is a technology for providing bi-directional full duplex communication channels over a single TCP socket.

Are a new feature in HTML5 (draft) that let you stream data to and from web browsers.

Websocket



With sockets, send data back and forth anytime:



MQTT

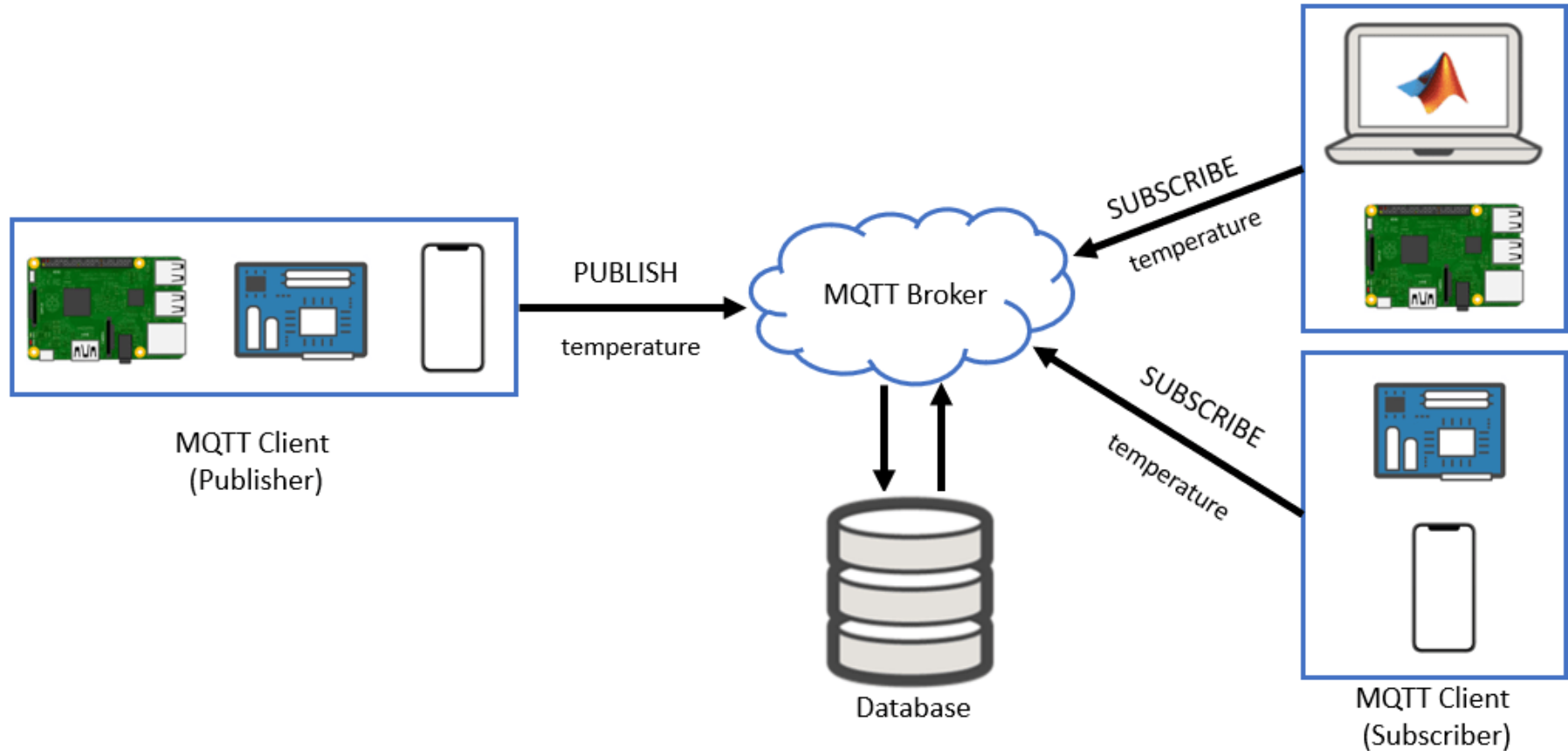
Message Queuing Telemetry Transport (MQTT) is a publish-subscribe architecture that is developed primarily to connect bandwidth and power-constrained devices over wireless networks. It is a simple and lightweight protocol that runs over TCP/IP sockets, WebSockets, and SSL (secure sockets layer).

MQTT has two components:

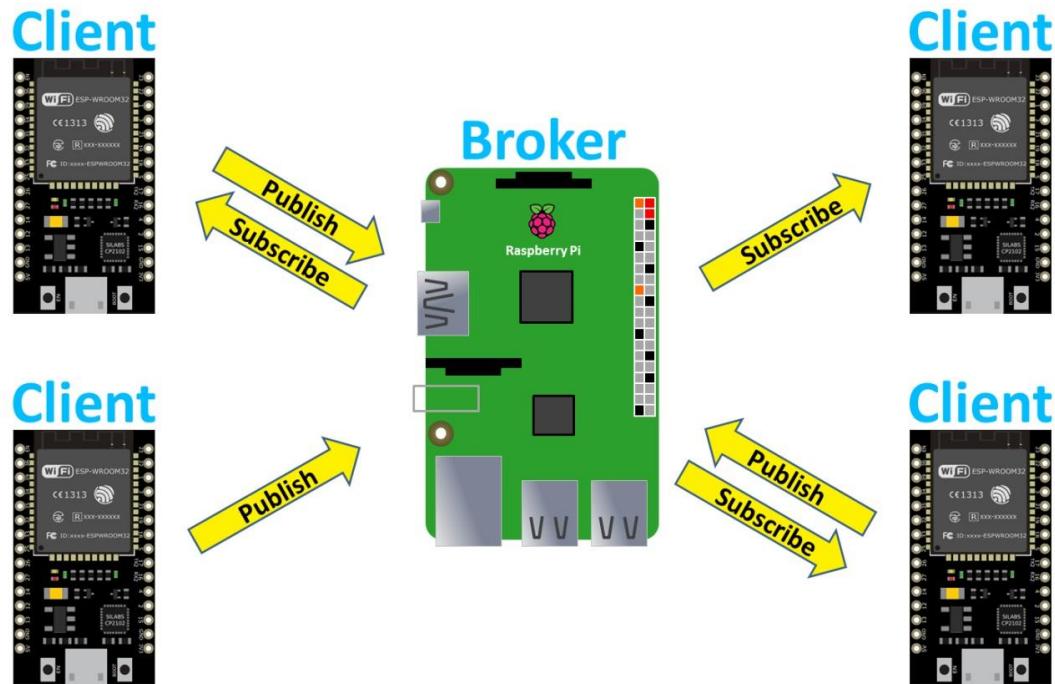
MQTT broker — An MQTT broker is a central point of communication. The broker is responsible for dispatching all messages between the clients.

MQTT client — An MQTT client is any device (for example, a computer or a mobile phone) that connects to the broker. A client that sends messages is a publisher. A client that receives messages is a subscriber. To receive a message, the client must subscribe to the topic of that message.

MQTT Diagram



Lab : MQTT Broker with Raspberry



Agenda

